Internship Report

Master in Competitive and Business Intelligence

Edge-Based Defense in Networks

A Modification of Bloch, Chatterjee & Dutta (2023)

with Applications to Sustainable Resource Management

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Abstract

This report studies what happens when defense in a network attack game is moved from the nodes to the connections between them. The starting point is the model by Bloch, Chatterjee and Dutta (2023), where each target invests in protection at its own location and an attacker chooses both a target and a path to reach it. The idea of placing defense on connections instead of locations comes from Mavronicolas et al. (2008), who showed that this small change produces equilibria with very different structure.

Keeping Bloch's setup of continuous investment and independent defenders, the report builds the modified model in full, derives the main results, and explains the reasoning behind each finding. The conclusions are illustrated using settings drawn from sustainable resource management — the illegal exploitation of shared natural resources across boundaries, such as transboundary logging, fishing in shared waters, and cross-border waste trafficking. An interactive R Shiny simulator was built to test the results numerically and to allow further experiments. The work connects two strands of the literature that have so far stayed apart: continuous-investment games on networks, and edge-based defense.

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General Introduction

The Setting

Many of the systems we depend on are networks, and many of them carry resources worth protecting. National parks and forests are connected by access corridors. Fishing grounds are reached through a limited set of maritime passages. Trade in goods — legal and illegal — moves along networks of roads, ports, and river routes. These networks are built to serve a purpose, but they share one uncomfortable feature: the resources they carry attract actors who want to exploit them, and those actors choose their routes deliberately.

Illegal logging, illegal fishing, poaching, and the trafficking of waste and protected species all follow this pattern. The people carrying out these activities are not random forces of nature; they are strategic. They study where enforcement is weak, choose the routes that offer the best return for the lowest chance of being caught, and shift their behavior when defenses change. Protecting shared natural resources means protecting the networks through which they are reached — and doing so against an adversary who is thinking just as carefully as the defender.

To make the picture concrete, consider three examples that recur throughout this report. In illegal logging across transboundary forests, the *attacker* is a logging operation moving illegally harvested timber out of the forest; the *defenders* are the park authorities or national agencies that patrol the access corridors — roads, trails, and river crossings — on each side of the boundary; the *value of each location* is the timber accessible from that location, which is what determines how attractive each forest region is to the logger. In illegal fishing in shared waters, the *attacker* is a fishing vessel operating without authorization; the *defenders* are the coast guards of the neighboring coastal states; the *value of each location* is the fishing ground itself — the richness of the stock that the vessel is trying to reach. In illegal waste trafficking, the *attacker* is a trafficker moving hazardous waste across borders; the *defenders* are the customs and environmental agencies on either side of each transport corridor; the *value of each location* is the disposal site — the place where the waste can be dumped most cheaply, which is what makes one destination more attractive than another to the trafficker.

These three examples share a feature that shapes everything that follows. The cost of getting from one place to another — the trafficker's transport cost, the logger's fuel and labor, the vessel's time at sea — is real in practice but is deliberately not part of the model studied here. The attacker bears no per-action cost in this formulation, an assumption inherited from the model that this report builds on. What the model captures is which valuable destinations are worth reaching and which routes are watched closely enough to deter attempts. The attacker's logistical expenses are a separate concern, set aside here so that the strategic structure of defense placement can be studied in isolation.

The push for sustainable resource management makes this question sharper. Resources are increasingly managed across administrative and national boundaries: forests that straddle borders, fish stocks shared between coastal states, transport corridors that pass through several jurisdictions. The benefits of coordinated management are real, but so is the challenge of defending a network whose connections are watched by several separate authorities at once.

The Question

How should defense be organized in such a network? Two basic answers exist in the academic literature. The first places defense on the nodes themselves — on the individual sites, facilities, or stocks that an intruder might target. Each location takes care of itself, and an intruder who reaches a defended location is intercepted there. The second places defense on the connections — on the corridors, passages, and routes through which an intruder must travel. The same connection can be watched from either end, and an intruder trying to use it must get past whatever defense both ends have put in place.

These two arrangements lead to very different strategic situations. In the first, each location decides on its own how much to defend. In the second, the two ends of any connection share something — and that sharing changes how each side thinks about its own choices. Do they cooperate? Do they free-ride on each other? Does it matter how many connections a location has?

This report studies the second arrangement. It builds on a recent model by Bloch, Chatterjee and Dutta (2023), which places defense on nodes, and asks what changes when defense is moved to connections instead. In their setup, the attacker is a single external node sitting outside the network, and an attack is a path that the attacker chooses through the network to reach a target. The inspiration for the move to edge-based defense comes from Mavronicolas et al. (2008), who studied a different network security game in which the attackers target nodes directly while the defender protects an edge; in their model, edge-based defense produces a recognizably different equilibrium structure.

What This Internship Did

The internship took place at CRESE, the economic research center at Université Marie et Louis Pasteur in Besançon, under the supervision of Professor Sylvain Béal. The work had three connected goals.

The first goal was to build the modified model in full. This meant deciding exactly how to translate every piece of Bloch's original formulation into the new setting where defense lives on connections. Some pieces carry over without change. Others change in ways that are obvious. A few change in ways that are not obvious at all, and the report works through each of these carefully.

The second goal was to derive the main theoretical results. The model needed to be shown to have an equilibrium, the equilibrium needed to be shown to be unique, and the conditions defining the equilibrium needed to be written out explicitly. From there, the report examines how the equilibrium responds to changes in the network: a target becoming more valuable, a new connection being added, or a target being removed from consideration. Each of these results is presented with the reasoning behind it explained as plainly as possible.

The third goal was to make the results accessible through a working tool. An R Shiny simulator was built that takes any network the user describes, computes the equilibrium under both defense models (Bloch's original node defense and the modified edge defense), and presents the comparison side by side. The simulator served two purposes: it allowed the theoretical results to be verified numerically on real examples, and it allowed comparative experiments to be run in seconds rather than hours.

The Central Problem

The central question this report addresses is the following: *when defense in a network is placed on the connections rather than the locations, how does the strategic situation change, and what does this tell us about the protection of shared resources?* The question is theoretical in form but practical in motivation. The routes through which shared natural resources are reached — forest corridors, maritime passages, transport networks — are monitored along their connections by separate authorities, not at single central points. Understanding how this changes the strategic balance between a strategic adversary and decentralized defenders is the contribution this work tries to make.

Structure of the Report

The report is organized into four chapters. Chapter 1 presents the research landscape: the broader literature on network security games, the specific tradition that Bloch's model belongs to, and the smaller body of work on edge-based defense. The chapter ends by identifying the gap that this work fills.

Chapter 2 develops the modified model under complete information. It begins with the practical motivation, sets up the notation, presents the similarities and differences from Bloch carefully, and works through the formal results. Numerical examples are used throughout to illustrate why each result holds. The chapter closes with a comparative section on how the management of a shared edge can be modelled in four ways, organised around a two-by-two table that crosses two modelling choices — whether the two endpoints' efforts combine independently or jointly, and whether their costs are paid separately or shared.

Chapter 3 extends the analysis to incomplete information. Two distinct kinds of uncertainty are treated in turn: private target values, where defenders know only their own valuation, and partial network observation, where each player observes only its own incident edges. Each case is developed as a Bayes-Nash equilibrium on the triangle network, with side-by-side comparisons to the complete-information baseline.

Chapter 4 covers the implementation. The R Shiny simulator is described, the numerical results are reported, and the chapter closes with a reflection on the skills developed during the internship and the lessons learned.

The general conclusion summarizes the findings, acknowledges the limits of the work, and suggests directions for future research.

Chapter 1

The Research Landscape

Introduction to the Chapter

This chapter places the report's contribution within the broader academic conversation about how networks should be defended against attacks. The literature on this question has grown in two distinct directions, which until now have stayed largely separate. One direction looks at how defenders allocate resources at the locations of a network. The other looks at how defenders interfere with the connections between locations. The model studied in this report sits between these two traditions: it borrows the continuous-investment framework from the first and the placement of defense from the second. To make the contribution clear, the chapter walks through both traditions, identifies what each one offers and what each one leaves out, and then states explicitly the gap that this work fills.

1.1 Network Security Games: The Field

A network security game is a simplified picture of a real situation in which one or more attackers try to do harm to a system represented as a network, and one or more defenders try to prevent that harm. The network is usually drawn as a set of points (called nodes) connected by lines (called edges). The attacker chooses a target — typically one of the nodes — and tries to reach it. The defender allocates protective resources somewhere in the network, hoping to stop the attack before it succeeds.

The reason such a simplified picture is worth studying is that it captures something real. Forest corridors, shared fishing grounds, cross-border trade routes, power grids, water distribution networks, and communication systems all have the property that they are made of locations linked by connections, that they carry something valuable, and that they are exposed to actors who want to exploit or disrupt them. The strategic question — how to protect a network when its defenders have limited resources and its attackers are intelligent — is the same across all of these settings, even when the technical details differ.

The field has produced a wide range of models, distinguished from one another by three main choices. First, who is the defender: a single central authority deciding for the whole network, or many independent defenders each protecting only their own piece? Second, what is the form of defense: an all-or-nothing decision to protect a location, or a continuous investment that gradually raises the chance of interception? Third, where is defense placed: at the nodes themselves, or on the connections between them?

A fourth choice, less often highlighted but equally important, concerns how value is distributed across the network: each node may carry its own fixed value, independent of the others, or value may emerge from the network structure itself (as in Dziubiński & Goyal 2017, where the value of a node depends on which others it disconnects, or Cerdeiro et al. 2017, where value comes from being connected to others). The model in this report follows the first convention, as in Bloch et al. (2023): each target has its own fixed value, independent of the values of other targets.

Different combinations of these three choices produce different models. The work in this report can be located by the choices it makes: the defenders are independent (decentralized), their investment is continuous, and their defense is placed on the connections. This combination is, to my knowledge, new. The chapter that follows shows what already exists in the surrounding literature.

1.2 Defense at the Nodes: The Centralized Tradition

A first body of work looks at defense placed at the nodes of a network, with a single centralized defender deciding on the whole protection plan. The reasoning is straightforward: the defender owns or coordinates the network, can see it as a whole, and chooses where to put protection so as to minimize the damage from a future attack. In this tradition, the defender bears the cost of protection.

Dziubiński and Goyal (2013) study a model in which the designer chooses both the network and the defense at the same time, against an attacker who is given a fixed budget for striking nodes. Both linking and defense are costly to the designer, while the attacker operates under a budget rather than a per-attack cost. Their results show that the optimal defense is sharply binary: either the network has zero defense, exactly one protected node, or every node protected. Intermediate solutions are rarely optimal. The paper produces a clear taxonomy of when each regime is best, depending on the cost of building connections and the cost of defending nodes.

Dziubiński and Goyal (2017) sharpen the analysis further by introducing the idea of a network separator — a set of nodes whose removal disconnects the network. In their formulation, both the defender and the attacker face per-action costs: the defender pays

c_D to protect each node, and the attacker pays c_A to attack each one. This more abstract reformulation produces some surprising results, including a version of Braess's paradox: adding a connection to a network can sometimes hurt the defender, because it creates a path that the attacker can exploit.

Cerdeiro et al. (2017) change one important assumption: instead of a centralized defender, each node decides on its own whether to protect itself, paying its own cost privately. The threat is modeled as exogenous exposure rather than a strategic player. The strategic situation is different from the centralized case because individual nodes do not internalize the benefit their protection gives to others. The paper measures the cost of decentralization and shows that, in many cases, it is severe — decentralized choice can lead to dramatic underinvestment in protection compared to the centralized optimum.

What unites these papers is that defense is placed on nodes and is typically all-or-nothing, with the defender always paying the cost of protection. The treatment of the attacker varies: a budget in some formulations, a per-action cost in others, and an exogenous threat in yet others. The strength of the tradition is that it produces clean, sharp characterizations of optimal network design and protection. Its limitation is that real defense is often not all-or-nothing — it is a question of degree — and is often paid for not at locations but along connections.

1.3 Continuous Defense at the Nodes: Bloch, Chatterjee & Dutta (2023)

Bloch et al. (2023) occupies a particular position in this literature. Like the papers above, defense is placed at the nodes. Unlike them, defense is continuous: each defender chooses how much to invest, with a strictly increasing and strictly convex cost function. The quadratic specification used throughout most of their analysis is a tractable choice from this broader class; the authors note explicitly that the main results do not depend on this specific form. And unlike the centralized tradition, the defenders are independent — each target node makes its own investment decision based on its own value and its own perception of the threat.

The model has several distinctive features. The attacker chooses both a target and a path to reach it, and so the attack itself moves through the network rather than landing instantly. Each node along the path may intercept the attack, with a probability that depends on its own investment. Targets have heterogeneous values, and each defender's payoff depends only on its own target's value — that is, defenders are independent in their concerns, even though the values themselves are common knowledge. The result is a richer game than the binary models above: the equilibrium involves not only which targets are attacked but also how much each defender invests, how the attacker spreads probability across paths, and how the network's topology shapes both decisions.

Bloch and his coauthors show that under mild conditions the game has a unique equilibrium. They characterize that equilibrium fully for several network shapes, including line networks, trees, and stars. They derive comparative statics: how the equilibrium shifts when a target's value changes, when a new connection is added, or when a target is dropped from the attacker's consideration. They also study a cooperative version of the model, in which all defenders coordinate their investment decisions. The cooperative arrangement makes the attacker strictly worse off than under decentralized defense, because coordinated defenders can engineer changes to the support that no individual defender would choose alone.

The model in Bloch et al. (2023) is the foundation on which this report builds. Its strength is the combination of continuous investment, decentralized choice, and a path-based attacker — features that together make the model both realistic and tractable. Its limitation, from the point of view of this report, is that defense remains located at the nodes. The question this report asks is what happens if that one feature is changed.

One computational question that Bloch et al. (2023) did not resolve was whether the unique Nash equilibrium of their game could be computed efficiently for an arbitrary network. Bloch's characterization is given in terms of the equilibrium support — the set of targets attacked with positive probability — and identifying this support for general networks was left as a problem for future work. Kaźmierowski and Dziubiński (2023) answer that question: they show that a Nash equilibrium of the Bloch model can be computed in polynomial time, specifically in $O(n^4)$ with respect to the number of nodes. Their algorithm proceeds by identifying a class of nodes called *linkers* — nodes that are never attacked in any equilibrium — and reducing the network by removing them, after which the support can be found by an iterative procedure that generalises Bloch's own characterization for line networks. This is a computational complement to Bloch's work rather than a separate model: the game is the same, and the contribution is showing that finding its equilibrium is tractable.

1.4 Defense at the Connections: The Edge Tradition

A second body of work, much smaller, places defense on the connections themselves. In these models, the defender does not protect individual nodes; instead, the defender controls or interferes with the edges through which an attacker must move.

Washburn and Wood (1995) present the earliest version of this idea. A single attacker tries to move through a network from a source to a destination; a single defender chooses one edge to inspect, hoping to catch the attacker on it. The game is repeated daily, and both players use mixed strategies. The authors show that the value of the game can be computed using the maximum-flow / minimum-cut structure of the network — a connection that turns out to be central to the entire literature on edge-based interdiction.

Israeli and Wood (2002) extend the same idea to a richer setting in which the defender can interdict multiple edges, subject to a budget. Each interdicted edge becomes longer or impassable, and the attacker — the follower in this version of the game — chooses the shortest path that remains. The paper develops an algorithmic approach (using a technique called covering decomposition) that performs much better than direct optimization. The structural intuition is striking: when the attacker has many alternative paths, the defender must block enough of them to eliminate all the cheap routes. This idea — that the defender's job is to "cover" all the paths the attacker could take — reappears in the edge-based modification studied in this report.

Zenklusen (2010) approaches the same question from a different angle, studying the computational complexity of edge interdiction on planar graphs — networks that can be drawn without crossings, like most real geographical networks. The paper develops algorithms that are tractable for these special cases and shows their limits for general networks. While not directly used in this report, the work is part of the same conceptual tradition: defense lives on edges, and the question is how to allocate it efficiently.

Mavronicolas et al. (2008) stands somewhat apart. The setup is closer in spirit to the present work, even though the technical details differ. Multiple attackers each choose a node, and a single defender chooses one edge to protect; defending an edge covers both of its endpoints simultaneously. The authors show that under suitable conditions the equilibrium has an elegant matching structure: the attacked nodes form an independent set (none of them adjacent), and the defender's chosen edges form a matching that pairs each non-attacked node to a unique attacked one. This conceptual idea — that defense on an edge produces coverage spillover between its endpoints — is the direct inspiration for the modification studied in this report.

The strength of the edge tradition is that it captures something the node tradition misses: defense is often paid for along connections, not at locations, and shared connections create strategic interactions that node-based models cannot represent. The limitation is that the edge tradition has so far worked mostly with discrete, all-or-nothing defense by a centralized defender. The combination of continuous investment, decentralized defense, and edge placement has not been studied.

1.5 The Gap

Two strands of literature, each with its own strengths and limitations, have grown in parallel. The node tradition has produced the most refined models of decentralized continuous defense, with Bloch et al. (2023) as its most developed example. The edge tradition has produced powerful results about the structure of optimal interdiction, but has stayed within discrete, centralized formulations.

What is missing is a model that combines the realism of continuous decentralized investment with the structural insight of edge-based defense. The present work is an attempt to fill this gap. It takes Bloch's framework as its starting point and modifies one feature: defense is moved from the nodes to the edges, with both endpoints of an edge able to invest independently. The modification is small to state but consequential to derive. It changes how survival probabilities are computed, how defenders interact with one another, how multi-path targets behave, and how cooperative defense compares to decentralized defense. Each of these consequences is the subject of the next chapter.

Conclusion to the Chapter

The literature on network security games has converged on a common set of questions but has answered them in two different ways. Defense at the nodes offers tractability and realism for individual locations; defense at the edges offers a clearer picture of how connections matter. Bloch et al. (2023) brings the node tradition to a high level of sophistication. Mavronicolas et al. (2008) shows what edge-based defense can look like in a discrete world. The present work bridges the two by asking what happens when continuous, decentralized defense is placed on edges — and, as the next chapter will show, the answer is not a small adjustment but a meaningful change in the strategic landscape.

Chapter 2

The Edge-Based Defense Model

Introduction to the Chapter

This chapter develops the contribution of the report. It begins with the practical motivation that justifies moving defense from locations to connections, drawing on examples from sustainable resource management. It then summarizes the original model of Bloch, Chatterjee and Dutta (2023) so the reader has a clear reference point. The modified model is built up step by step, with every piece of notation introduced once and used freely afterward. After that, the chapter walks through the similarities with Bloch's model and then the differences, explaining the reasoning behind each change as plainly as possible. Two worked examples illustrate the main effects. The chapter closes with a discussion of how the management of each shared edge by its two endpoints can be modelled, organised around a two-by-two table of possibilities.

The mathematical content of this chapter is presented through formal statements, each followed immediately by its proof and then by a plain-language explanation of what the result means. Detailed step-by-step calculations for the numerical examples are collected in the Appendix.

2.1 The Practical Motivation

The model studied in this chapter has two features that any motivating example must respect. First, the attacker is strategic: it chooses where to strike and which route to take, weighing the value of each target against the chance of being intercepted. Second, defense is decentralized and placed on connections: the two ends of any connection are separate decision-makers, each investing in monitoring the connection they share. A real-world motivation is only convincing if it has both a genuine strategic adversary and connections that two distinct defenders both have reason to watch.

The natural setting for such examples, within sustainable resource management, is the illegal exploitation of shared natural resources across administrative or national boundaries. In each of the cases below, the resource is depleted by actors who behave strategically, and the routes they use are monitored jointly by separate authorities.

Consider illegal logging in a transboundary forest. Two countries, or two regional park authorities, each manage adjacent stretches of forest connected by a small number of access corridors — roads, trails, and river crossings through which timber must be moved out. Illegal loggers are strategic: they choose which corridor to use based on the value of the timber it gives access to and the chance of being caught along the way. Each authority can fund patrols, install camera traps, or run checkpoints along the corridors on its side. A corridor that runs along or across the boundary is watched by both authorities at once, and each authority's effort on that corridor affects the other's incentive to invest.

Illegal fishing in shared waters has the same structure. Two coastal states manage adjacent maritime zones connected by a limited set of passages — straits, channels, and established routes through which fishing vessels enter and leave. Vessels fishing illegally choose their entry routes strategically, balancing the richness of the fishing grounds a route opens up against the patrol presence along it. Each state invests in coast guard patrols and surveillance along the passages it shares with its neighbor. The passage is the connection; both states have reason to monitor it.

Illegal waste trafficking completes the picture. Hazardous waste is moved illegally across borders to be dumped where regulation is weakest, travelling along a transport network of highways, ports, and river routes. The target's value here is the disposal site itself — the place where the waste can be dumped at lowest regulatory cost — which is what makes one destination more attractive to the trafficker than another. The traffickers are strategic, routing shipments through whichever corridor offers the weakest combined inspection. Customs and environmental agencies on either side of a shared corridor each invest in inspections, and the corridor between them is defended jointly.

These three examples share the feature that the original Bloch model does not capture. Defense is paid for along connections, not at locations, and the two ends of any connection are separate authorities that share both the threat and the cost. The strategic situation that arises from this sharing — whether the two ends cooperate, whether they free-ride on each other, whether the structure of the network amplifies or dampens these effects — is the question the rest of this chapter addresses.

2.2 Bloch's Original Model

Before introducing the modification, we summarize the model of Bloch, Chatterjee and Dutta (2023). The reader who is already familiar with the paper can skim this section; for everyone else, it provides the reference point against which the modified model is built.

The model is played on a network $G = (V, E)$ where $V = \{0, 1, 2, \ldots, n\}$ is a set of nodes and E is a set of edges. Node 0 is the attacker. Nodes $1, \ldots, n$ are potential targets. Each target i has a value $b_i \in (0,1)$, representing the damage caused if a successful attack reaches it. Values are assumed generic, meaning $b_i \neq b_j$ for all $i \neq j$.

The attacker chooses a probability distribution π over the set of paths $\mathcal{P}$ from node 0 to target nodes. Each path p terminates at some target $k(p)$. By summing the probabilities of all paths ending at the same target, the distribution over paths π induces a distribution q over targets: $q_i = \sum_{p:k(p)=i} \pi(p)$.

Each target i chooses a single interception investment $x_i \in [0,1]$. This investment has a cost of $x_i^2/2$. When an attacker passes through node i on the way to its destination, the probability that node i intercepts the attacker is x_i. Equivalently, the probability that the attacker survives passage through node i is $(1 - x_i)$. The investment x_i applies regardless of which direction the attacker comes from — this is the central feature of node defense.

The probability that the attack survives an entire path p to reach the target $k(p)$ is the product of survival probabilities along the path:

$$\alpha_{k(p)}(p) = \prod_{\substack{j \in V(p) \\ j \neq 0,\, j \neq k(p)}} (1 - x_j),$$

where $V(p)$ is the set of nodes on path p. The probability of a successful attack on $k(p)$ is then $\beta_{k(p)}(p) = \alpha_{k(p)}(p) \cdot (1 - x_{k(p)})$, with the additional factor accounting for interception at the target itself.

The attacker's expected payoff is:

$$U(\pi, x) = \sum_{p \in \mathcal{P}} \pi(p) \cdot \beta_{k(p)}(p) \cdot b_{k(p)}.$$

The attacker bears no cost of attacking in this model; it simply maximizes expected damage, with nothing subtracted. This is a simplifying assumption — one that a natural extension of the model would relax by giving the attacker a per-attack or per-path cost — but it is the baseline in Bloch's formulation, and it is worth stating explicitly because the defender's payoff just below does carry a cost term.

Each defender i has the payoff:

$$V_i(\pi, x) = -\sum_{\substack{p \in \mathcal{P} \\ k(p)=i}} \pi(p) \cdot \beta_i(p) \cdot b_i \;-\; \frac{x_i^2}{2}.$$

The first term is the expected loss from attacks targeting node i; the second is the cost of investment. Bloch and his coauthors note that the quadratic form of this cost is a simplifying choice: their main results go through for any strictly increasing, strictly convex cost function. The quadratic form is retained here for the same reason — tractability —

and the modified model inherits it.

A notational convention used throughout the rest of the chapter: in equilibrium we write simply U for the common equilibrium value $U(\pi^*, x^*)$ — the single number that, by Lemma 2.6, every attacked target yields. Every appearance of a bare U in the analysis below refers to this equilibrium value.

Bloch and his coauthors prove existence and uniqueness of the Nash equilibrium under the genericity assumption. To make the comparison in the next section concrete, it is worth setting out the structure of that equilibrium, because the modified model changes specific pieces of it.

Bloch distinguishes two kinds of target. A *first target* is one the attacker can reach directly from node 0 along a path whose intermediate nodes invest nothing. A *deeper target* is one reached only by passing through other targets that are themselves attacked. For a first target i, Bloch's equilibrium gives

$$x_i = 1 - \frac{U}{b_i}, \qquad q_i = \frac{1}{b_i}\left(1 - \frac{U}{b_i}\right),$$

and for a deeper target i with immediate predecessor k,

$$x_i = 1 - \frac{b_k}{b_i}, \qquad q_i = \frac{b_k}{b_i U}\left(1 - \frac{b_k}{b_i}\right).$$

These equilibrium equations come from two ingredients: the attacker's indifference across all attacked targets (every target in the support yields the same payoff U), and each defender's first-order condition (each defender invests up to the point where the marginal cost of investment equals the marginal reduction in expected loss). The condition $\sum_i q_i = 1$ then pins down U. Because each q_i is a strictly decreasing function of U, this equation has a unique solution.

Bloch also studies a cooperative version of the game, in which all defenders coordinate their investment decisions to minimize total expected loss. The cooperative arrangement makes the attacker strictly worse off than under decentralized defense, because coordinated defenders can engineer changes to the set of attacked targets that no individual defender, acting alone, would choose.

The defining feature of the model, for the purposes of this chapter, is that defense is placed at nodes. A single investment x_i defends node i from *any* direction of approach. The next section moves this defense to the edges and works out what changes.

2.3 The Edge-Based Modification

We now describe the modified model. The network, the attacker, the target values, and the genericity assumption are all retained from Bloch's formulation. What changes is how defenders invest and how interception occurs.

Definition 2.1 (Edge investment). *Each defender $i \in \{1, \dots, n\}$ chooses a vector of investments $(x_{i,e})_{e \in E(i)}$, where $E(i) = \{e \in E : i \in e\}$ denotes the set of edges incident to node i. Each investment $x_{i,e} \in [0, 1]$ is chosen independently and represents the interception effort that node i places on edge e.*

In words, each node is no longer choosing a single defense level; instead, it chooses a separate level for each connection it participates in. This models decentralized self-defense: a node allocates protective resources across the connections it shares with others, possibly investing more on connections it perceives as more vulnerable.

Definition 2.2 (Edge survival probability). *For an edge $e = (a, b) \in E$, both endpoints may invest. The probability that an attacker survives the edge is:*

$$S(e) = (1 - x_{a,e})(1 - x_{b,e}).$$

The interpretation is that each endpoint independently attempts interception, and the attacker survives the edge only if both attempts fail. This product structure captures the idea that defense on a shared connection compounds: even modest investments from both sides can produce substantial total protection.

This is one natural way to combine the two endpoints' efforts, but it is not the only one. An alternative is the *joint* survival probability

$$S(e) = 1 - x_{a,e}\, x_{b,e},$$

under which interception requires both endpoints to act: the attacker is caught only with probability equal to the product of the two efforts. The two formulations describe genuinely different interception technologies — independent attempts versus joint action — and Section 2.7 returns to compare them. For the main analysis of this chapter we adopt the independent-attempts form of Definition 2.2.

A clarification is needed about when the formula is genuinely two-sided. If only one endpoint invests — because the other has no incentive to do so — then the formula collapses to the original Bloch survival expression. In particular, for the entrance edge $e_1 = (0, b_1)$ between the attacker and the first node, the attacker invests nothing, so $x_{0,e_1} = 0$ and $S(e_1) = (1 - x_{b_1,e_1})$, which is exactly Bloch's expression.

Definition 2.3 (Path survival probability). *For a path $p = (e_1, e_2, \dots, e_L)$ from node 0 to a target $k(p)$, the probability that the attacker survives the entire path is the product of edge survival probabilities:*

$$\alpha_{k(p)}(p) = \prod_{\ell=1}^{L} S(e_\ell) = \prod_{\ell=1}^{L} (1 - x_{a_\ell,e_\ell})(1 - x_{b_\ell,e_\ell}),$$

where $e_\ell = (a_\ell, b_\ell)$ is the ℓ-th edge on the path.

In Bloch's model, the path survival probability α and the success probability β are distinct: the attacker first survives the intermediate nodes (factor α) and then survives the target's own defense (factor $1 - x_{k(p)}$). In the edge-based model, the target's investment is already absorbed into the survival product, because the last edge on the path includes the target as one of its endpoints. There is no separate interception step at the target. Therefore $\alpha_{k(p)}(p) = \beta_{k(p)}(p)$ in this model.

Definition 2.4 (Defender cost)**.** *The total cost incurred by defender i is:*

$$C_i = \sum_{e \in E(i)} \frac{x_{i,e}^2}{2}.$$

Each edge investment incurs its own quadratic cost. This cost structure is *separable* in two distinct senses, and it is worth keeping them apart. First, it is separable *across the edges of a single defender*: defender i's cost on one of its incident edges does not affect its marginal cost on another. Second, it is separable *within an edge, between the two endpoints*: on a shared edge $e = (i, j)$, defender i's cost depends only on its own investment $x_{i,e}$, not on the neighbor's investment $x_{j,e}$. The first kind of separability means that defender i's first-order condition for each of its edges can be solved independently of its other edges. The second kind — separability between endpoints — is the assumption that Section 2.7 relaxes when it considers a shared cost on each edge.

The attacker's payoff is:

$$U(\pi, x) = \sum_{p \in \mathcal{P}} \pi(p) \cdot \alpha_{k(p)}(p) \cdot b_{k(p)},$$

and each defender's payoff is:

$$V_i(\pi, x) = -\sum_{\substack{p \in \mathcal{P} \\ k(p)=i}} \pi(p) \cdot \alpha_i(p) \cdot b_i \; - \sum_{e \in E(i)} \frac{x_{i,e}^2}{2}.$$

As in Bloch's model, the attacker bears no cost. Both payoffs are written explicitly as functions of π (the attacker's mixed strategy) and x (the full profile of defender investments). It is worth noting that only π is mixed; defenders choose their investments deterministically. This is without loss of generality: because each defender's payoff is concave in its own investment vector (the loss term is linear in each $x_{i,e}$ and the cost term is strictly concave when the payoff is written as a quantity to be maximized), a defender can never do strictly better by randomizing over investment levels than by playing the best deterministic level. Checking pure deviations for defenders is therefore sufficient.

For the cooperative version of the model, a single coordinator controls all edge investments and minimizes the total expected loss across the network. Writing $\mathcal{P}_i$ for the set of paths ending at target i, the loss function is

$$\mathcal{L} = \sum_{i \in \Delta} \sum_{p \in \mathcal{P}_i} \pi(p) \left[\prod_{e \in p} (1 - y_{a_e, e})(1 - y_{b_e, e}) \right] b_i \ + \sum_{e \in E} \sum_{j \in e} \frac{y_{j,e}^2}{2},$$

where Δ denotes the set of attacked targets and $y_{j,e}$ the coordinator's chosen investment by node j on edge e. The outer double sum in the loss term runs over targets and then over the paths leading to each target: a single target can be attacked along several paths, and each contributes its own survival term. The double sum in the cost term reflects that every edge is paid for from both sides.

A note on terminology used throughout the rest of the chapter. When we speak of the *support*, we mean the set of targets attacked with positive probability — that is, the support of the induced distribution q over targets, written Δ. This is distinct from the support of π over paths. The two are related but not identical, and the equilibrium analysis is organized around the support over targets.

2.4 Similarities with Bloch

Several elements of Bloch's analysis carry over to the modified model, because they depend on properties of the attacker's choice rather than on where defense is placed. We collect them here, each with its proof.

Existence

Theorem 2.5 (Existence). *The edge-based defense game admits a Nash equilibrium in which the attacker plays a mixed strategy over paths and each defender plays a pure strategy over investments.*

Proof. The proof applies the Debreu–Fan–Glicksberg fixed-point theorem, which requires three conditions: compact and convex strategy spaces, continuous payoffs, and quasi-concavity of each player's payoff in its own strategy.

The attacker's strategy space is the set of probability distributions over the finite set of paths $\mathcal{P}$, which is the standard simplex — non-empty, compact, and convex. Each defender i's strategy space is $[0, 1]^{|E(i)|}$, the unit hypercube of dimension equal to the number of edges incident to i, which is also non-empty, compact, and convex.

The attacker's payoff $U(\pi, x)$ is a polynomial in the entries of π and x, hence continuous in (π, x). Each defender's payoff V_i is likewise a polynomial, hence continuous in (π, x_i).

For quasi-concavity: the attacker's payoff is linear in π for fixed x, hence quasi-concave in the attacker's own strategy. For defender i, fix π and the other defenders' investments

x_{-i}. The loss term of V_i is linear in each $x_{i,e}$, because the survival products are multilinear in the investments. The cost term $-\sum_e x_{i,e}^2/2$ is strictly concave. The sum is concave in x_i, hence quasi-concave.

All three conditions hold, so a Nash equilibrium exists. Because each defender's payoff is concave in its own investment, the best response of each defender is a pure investment vector; no defender can gain by randomizing. The equilibrium therefore has the attacker mixing over paths and each defender playing a pure investment profile. □

Lemmas Carrying Over

Three of the four key lemmas in Bloch's analysis depend only on the attacker's optimization and carry over. We state and prove each, indicating in the title the corresponding Bloch result.

Lemma 2.6 (Equal payoff across the support; Bloch's Lemma 1)**.** *In equilibrium, all targets in the support yield the attacker the same expected payoff U. Moreover, every path the attacker uses with positive probability yields the same payoff U.*

Proof. Suppose two targets i and j in the support gave the attacker different expected payoffs, with i strictly better. In a general network the attacker has separate paths leading to i and to j, and it could increase its expected utility by transferring probability mass from a path leading to j onto a path leading to i. This contradicts the assumption that the attacker's strategy is a best response. Hence all supported targets yield the same payoff U.

The same argument applies one level down. If two paths leading to the same target i yielded different payoffs, the attacker would shift mass from the worse path to the better one, so every path used with positive probability yields U. On a line network, where there is only one route through the network, the argument specializes: there are no parallel paths to transfer mass between, but the attacker still chooses how much probability to place on terminating the attack at each target along the single spine, and equal payoff across the support is the condition that no such reallocation is profitable. This rests on the values being strictly increasing along the path (Lemma 2.8 below, the carried-over Bloch Lemma 4(i)), which is what makes the cumulative survival decay precisely offset by the rising target value.

This path-level indifference, not just the target-level version, is what the mutual-coverage result of the next section relies on. □

Lemma 2.7 (Optimal predecessor has lowest value; Bloch's Lemma 3)**.** *If a target i can be reached through several support members acting as predecessors, the attacker's equilibrium route to i passes through the predecessor with the lowest value.*

Proof. Let j and k be two support members, each of which can serve as a predecessor of i — that is, from each there is a path to i that does not pass through any other support member. Both j and k are themselves in the support, so by Lemma 2.6 the attacker's payoff from attacking j equals its payoff from attacking k:

$$\alpha_j\, b_j = U = \alpha_k\, b_k,$$

where α_j and α_k are the survival probabilities of reaching j and k respectively. This single pair of equalities already absorbs everything that happens *before* j and before k: whatever defense was accumulated along the way to each predecessor, the equilibrium indifference condition has compressed it into the statement that both predecessors are worth exactly U to the attacker. It does not matter whether j is a first target with nothing accumulated before it, or a deeper target reached only after passing other support members — the equality $\alpha_j b_j = \alpha_k b_k$ holds either way.

Now suppose $b_j < b_k$. From $\alpha_j b_j = \alpha_k b_k$ it follows that $\alpha_j > \alpha_k$: the lower-value predecessor is reached with higher probability. To continue from a predecessor to i, the attack must survive that predecessor's defense on the connecting edge. A lower-value predecessor invests less in its own defense, so the attack survives it with higher probability, and combined with the higher arrival probability α_j, routing through j delivers a strictly higher probability of reaching i. The attacker therefore routes through the lowest-value predecessor.

The point worth emphasizing is the division of labor between the two lemmas. Lemma 2.6 is the *input*: it tells us every predecessor is worth U, which is what makes the comparison of predecessors clean. The present lemma is the *conclusion*: it turns that fact into a statement about which predecessor is the best gateway onward to a deeper target. The two questions — which predecessor is worth U, and which predecessor is the best route to i — are different, and the second is the one that needs proving. □

Lemma 2.8 (Values increase along attack paths; Bloch's Lemma 4(i))**.** *Let $i_1, i_2, \ldots, i_m$ be the support members on an attack path, in the order the attacker meets them. Then $b_{i_1} < b_{i_2} < \cdots < b_{i_m}$.*

Proof. Consider two consecutive support members i_ℓ and $i_{\ell+1}$ on the path. The attacker's payoff from attacking $i_{\ell+1}$ along this path is $\alpha_{i_{\ell+1}} b_{i_{\ell+1}}$, and from attacking i_ℓ along the same path truncated at i_ℓ is $\alpha_{i_\ell} b_{i_\ell}$. By Lemma 2.6, both equal U. The attack reaching $i_{\ell+1}$ has had to survive strictly more defense than the attack reaching i_ℓ — at least the defense on the edge between them — so $\alpha_{i_{\ell+1}} < \alpha_{i_\ell}$. For the products $\alpha_{i_\ell} b_{i_\ell}$ and $\alpha_{i_{\ell+1}} b_{i_{\ell+1}}$ to be equal, it must therefore be that $b_{i_{\ell+1}} > b_{i_\ell}$. The values increase strictly along the path. □

Lemma 2.9 (Higher-value reachable target must be in the support; Bloch's Lemma 4(ii)). *Let i be a target in the support, and suppose there is a path from i to some target j that does not pass through any other support member. If $b_j > b_i$, then j is also in the support.*

Proof. Suppose for contradiction that $b_j > b_i$ and j is not in the support. The attacker can consider a deviation: attack j along the path from the attacker that goes to i and then continues to j along the unprotected continuation. Let α_i be the equilibrium survival of reaching i, so $\alpha_i b_i = U$ by Lemma 2.6. The continuation from i to j crosses no other support member, and any non-support target on that continuation invests zero, so survival is unchanged across the continuation. The deviation payoff is therefore $\alpha_i b_j > \alpha_i b_i = U$, strictly above the equilibrium payoff. This contradicts the attacker playing a best response unless j is itself in the support. □

Uniqueness

Theorem 2.10 (Uniqueness). *Under the genericity assumption ($b_i \neq b_j$ for all $i \neq j$), the edge-based defense game has a unique Nash equilibrium.*

Proof. The argument follows the contradiction structure of Bloch et al. (2023, Theorem 2), with survival probabilities replaced by their edge-based analogs. Suppose two distinct equilibria exist, with supports Δ_1 and Δ_2 and attacker payoffs U_1 and U_2.

Step 1: equal supports give a unique payoff. If $\Delta_1 = \Delta_2$, the equilibrium equations (Propositions 2.14, 2.15, and 2.16) express each q_i as a strictly decreasing function of U, so the constraint $\sum_i q_i(U) = 1$ has at most one solution. Then $U_1 = U_2$, and the equilibria coincide. So if the equilibria are distinct, $\Delta_1 \neq \Delta_2$.

Step 2: one support contains the other. Suppose neither support contains the other, so there exist $i \in \Delta_1 \setminus \Delta_2$ and $j \in \Delta_2 \setminus \Delta_1$. We show $U_1 < U_2$ from the existence of i; the symmetric argument applied to j then gives $U_2 < U_1$, a contradiction.

Target i is attacked in equilibrium 1, so $b_i > U_1$. We need to show $b_i \leq U_2$, which together with the previous inequality gives $U_1 < b_i \leq U_2$ and hence $U_1 < U_2$. Two cases arise, following Bloch.

Case A: there exists a path from the attacker to i that does not intersect Δ_2. Every node on such a path is outside the support of equilibrium 2 and therefore invests zero on every edge of the path. The deviation attacking i along this path is undefended, with payoff b_i. Since i is not attacked in equilibrium 2, this deviation cannot exceed the equilibrium payoff: $b_i \leq U_2$, as required.

Case B: every path from the attacker to i intersects Δ_2. In particular, equilibrium 1's own attack path p to i intersects Δ_2. Let ℓ be the last point of Δ_2 encountered along p before reaching i. By Lemma 2.8 applied to equilibrium 1, the support members on p have strictly increasing values, so $b_\ell < b_i$. The segment of p from ℓ to i does not pass through any other member of Δ_2, by the choice of ℓ. Now apply Lemma 2.9 to equilibrium 2:

$\ell \in \Delta_2$, there is a path from ℓ to i avoiding Δ_2, and $b_i > b_\ell$, so i must be in Δ_2. This contradicts $i \in \Delta_1 \setminus \Delta_2$.

In both cases we have either $b_i \leq U_2$ (Case A) or an immediate contradiction (Case B). Case B itself rules out the assumption $i \in \Delta_1 \setminus \Delta_2$ in the presence of cycles around Δ_2, so what remains is Case A, which gives $U_1 < U_2$. The symmetric argument applied to j gives $U_2 < U_1$, a contradiction. So one support strictly contains the other; without loss of generality $\Delta_2 \subset \Delta_1$.

Step 3: the smaller support gives the higher payoff. With $\Delta_2 \subset \Delta_1$ strictly, the constraint $\sum_{i \in \Delta_2} q_i(U_2) = 1$ has fewer positive terms than $\sum_{i \in \Delta_1} q_i(U_1) = 1$. Since each q_i is decreasing in U, fewer terms summing to one requires a larger U, so $U_2 > U_1$.

Step 4: the smaller-support equilibrium fails the best-response test. Take any $i \in \Delta_1 \setminus \Delta_2$. The same two-case argument as in Step 2 applies: either there is a path from the attacker to i avoiding Δ_2, yielding deviation payoff $b_i > U_1$ that the smaller-support equilibrium cannot accommodate, or every path crosses Δ_2 and Lemma 2.9 forces $i \in \Delta_2$ — a contradiction with $i \notin \Delta_2$. The boundary case $b_i = U_2$ is ruled out by the genericity assumption that target values are pairwise distinct (Bloch's Assumption 1, inherited in Section 2.3). The equilibrium payoff U_2 is determined by the equilibrium equations as a smooth function of the value vector $(b_1, \ldots, b_n)$, so the equality $b_i = U_2$ would correspond to the value vector lying on a lower-dimensional surface in the space of possible value profiles — a measure-zero coincidence that the genericity assumption excludes. Bloch et al. (2023) handle the analogous boundary case in their model in the same way.

The supports must therefore coincide, and by Step 1 the equilibrium is unique. □

2.5 Differences from Bloch

The interesting changes happen elsewhere. We now describe each of them.

From One Variable to Many

In Bloch's model, defender i chooses a single investment x_i. In the edge-based model, defender i chooses a vector of investments, one per incident edge. The dimensionality of each defender's strategy space grows from 1 to $|E(i)|$, the degree of node i in the network. This change is mechanical, but it has consequences for everything that follows.

Two Interception Attempts Per Edge

Survival through a single edge is now $(1 - x_{a,e})(1 - x_{b,e})$ rather than $(1 - x_i)$. When both endpoints have an incentive to invest, both attempt interception independently, and the attacker has to escape both. This is the source of the new strategic interactions: the two ends of an edge now appear in each other's first-order conditions, and their choices are no longer independent.

Separable Cost Across Edges

In Bloch, defender i's cost is a single quadratic term $x_i^2/2$. In the edge-based model, it is a sum of independent quadratic terms across all incident edges. The convexity of each term is preserved, but the cost is now distributed across multiple decisions rather than concentrated in one. This separability is what makes the cost savings result possible in the cooperative analysis.

Modified Lemma 2: Mutual Coverage

Bloch's Lemma 2 states that, without loss of generality, the attacker uses exactly one path to reach each target in the support. The proof rests on the fact that, in the node-based model, multiple paths to the same target collapse to equivalent survival probabilities — the target's single investment covers all directions of approach, so the attacker is indifferent between routes.

This argument breaks down in the edge-based model, and its replacement is the following.

Definition 2.11 (Free-highway edge). *An edge e incident to target i is a* free-highway edge *to i if there is a path from node 0 to i whose last edge is e and whose intermediate nodes each invest zero on every edge of that path. In other words, the attacker can reach i through e along a route that meets no positive defense until the edge e itself.*

Notation (multi-path factor m_i). For a first target i, define

$$m_i = |\{e \in E(i) : e \text{ is a free-highway edge to } i\}|.$$

Lemma 2.12 (Mutual coverage on free-highway edges). *Let i be a first target in the support with m_i free-highway incoming edges. In equilibrium the attacker spreads its attack on i across all m_i of these edges, and the defender invests an equal amount on each.*

Proof. Suppose, for contradiction, that the defender of i invested different amounts on two free-highway incoming edges e_1 and e_2, say $x_{i,e_1} > x_{i,e_2}$. The attacker's payoff from reaching i through e_2 would then be $b_i(1-x_{i,e_2})$, strictly larger than the payoff $b_i(1-x_{i,e_1})$ through e_1. By Lemma 2.6, every path the attacker uses must yield the same payoff, so the attacker would place no weight on e_1 and shift everything to e_2. But then e_1 carries no attack probability, and the defender's best response on e_1 is to invest nothing — contradicting $x_{i,e_1} > x_{i,e_2} \geq 0$ with e_1 defended. The same reasoning rules out the attacker placing unequal weight on the two edges while the defender responds optimally. The only configuration consistent with both sides best-responding is equal attack weight and equal defensive investment across all m_i free-highway edges. □

This is the continuous analog of the mutual coverage condition in Mavronicolas et al. (2008): the defender must cover every edge that could be attacked, and the attacker must threaten every edge that is defended; otherwise one side could improve.

A clarification is essential about what m_i counts. The factor m_i counts the number of *distinct edges incident to the target* that are free-highway edges, not the number of paths from the attacker. If several paths from node 0 converge onto a single final edge before reaching i, all of them share that one final edge, and together they contribute 1 to m_i, not more. The structural weakness comes from having to defend several distinct incoming edges, each needing its own investment; it does not come from the mere existence of multiple route options upstream that funnel into the same edge. The factor m_i therefore ranges from 1 for a target with a single free-highway incoming edge up to the number of distinct free-highway edges the target actually has.

Pure Nash Equilibrium: An Extra Structural Requirement

The same mutual-coverage logic that drives Modified Lemma 2 also alters the conditions under which a pure Nash equilibrium can be sustained. The algebraic conditions on the values are inherited unchanged from Bloch; the change is structural, an extra requirement that did not exist in the node-based model.

Proposition 2.13 (Pure NE conditions)**.** *A pure Nash equilibrium exists if and only if there is a target i^* such that*

(i) $b_{i^*}(1 - b_{i^*}) \geq b_j$ *for every target $j \neq i^*$ such that some path from 0 to j does not pass through i^*;*

(ii) $b_{i^*} \geq b_j$ *for every target $j \neq i^*$ such that every path from 0 to j passes through i^*;*

(iii) *the network (V, E) contains no cycle through both 0 and i^* — equivalently, i^* is reachable from 0 by a single (effective) attack path.*

Proof. Suppose the attacker plays a pure strategy concentrating on target i^* along a single path p^*. The defender of i^* then faces attacks arriving only along p^*, and its only incentive to invest is on the edges of p^*; in equilibrium it places investment on the final edge of p^* so that the attacker's success probability is $1 - b_{i^*}$, giving the attacker payoff $b_{i^*}(1 - b_{i^*})$. Every other defender, attacked with probability zero, invests nothing.

The attacker has no profitable deviation to a target j reachable by a path avoiding i^*: such a deviation meets zero defense and yields b_j, so we need $b_{i^*}(1 - b_{i^*}) \geq b_j$, which is condition (i). For a target j reachable only through i^*, deviating yields $b_j(1-b_{i^*})$, so we need $b_{i^*}(1-b_{i^*}) \geq b_j(1-b_{i^*})$, that is $b_{i^*} \geq b_j$, which is condition (ii). Conversely, if (i) and (ii) hold the described profile is a mutual best response *along the chosen path.*

Condition (iii) is the genuinely new content. Suppose the network contained a cycle through both 0 and i^*. Walking the cycle in two directions gives two distinct paths from 0 to i^* that enter i^* through different edges, so i^* has at least two incoming free-highway edges. If the defender of i^* concentrated its investment on the one edge the attacker is using, the other incoming edge would be left undefended, and the attacker would have a profitable deviation: switch to the undefended parallel edge and reach i^* with higher survival. So a pure equilibrium in which the defender invests on a single edge can be sustained only when the network has no cycle through 0 and i^* — equivalently, when i^* is reachable from 0 by a single effective attack path. This requirement does not exist in Bloch's node-based model, where a single investment x_{i^*} covers the target from every direction of approach. □

As a concrete instance, consider the line ATK–T_1–T_2 with values $b_1 = 0.10$ and $b_2 = 0.50$. Condition (i) is satisfied, since $b_2(1-b_2) = 0.50 \times 0.50 = 0.25 \geq 0.10 = b_1$, and condition (ii) is vacuous because no target lies behind T_2. Condition (iii) holds trivially: the network is a line, with no cycles. The game therefore has a pure equilibrium: the attacker attacks T_2 with probability one, T_2 invests $x_2 = b_2 = 0.50$ on its single incoming edge, T_1 invests nothing, and the attacker's payoff is $U = b_2(1-b_2) = 0.25$. The simulator reproduces these values exactly. Under cooperative defense on the same network the attacker's payoff falls to $U \approx 0.196$, consistent with the general principle that coordinated defense leaves the attacker weakly worse off; the full cooperative calculation is given in the Appendix.

To see what condition (iii) actually does, modify the network by adding a direct edge from the attacker to T_2, so that the network is now the triangle ATK–T_1–T_2 together with ATK–T_2. The values are unchanged: $b_1 = 0.10$, $b_2 = 0.50$. Conditions (i) and (ii) still hold: $b_2(1-b_2) = 0.25 \geq 0.10 = b_1$ as before, and (ii) is vacuous. But condition (iii) now fails: the triangle ATK–T_1–T_2–0 is a cycle containing both 0 and $i^* = T_2$, so T_2 has two distinct free-highway incoming edges. Whichever single edge T_2 defends, the attacker switches to the undefended one and reaches T_2 with full survival, yielding $b_2 = 0.50 > 0.25$. No pure equilibrium exists in this network; the only equilibrium is mixed, with T_2 investing equally on both incoming edges and the attacker spreading attack mass across them.

First-Target and Deeper-Target Equilibrium Equations

Before stating the equilibrium equations, a note on how a target's incoming routes can be heterogeneous in this model. In Bloch's node-based model, Bloch's Lemma 2 forces the attacker to reach each target along a single path, so a target sits at the end of exactly one incoming route in equilibrium — either a free-highway route or a route through an in-support predecessor, but never both. The edge-based model is different. Modified Lemma 2.12 forces the attacker to threaten every free-highway incoming edge of a first target, so a target genuinely receives multiple incoming routes at once; and when some of those routes are free-highway while others come from in-support predecessors, the target's incoming routes are heterogeneous.

We address this in stages. The next two propositions treat the homogeneous cases — a target whose incoming edges are all free-highway (first-target case), and a target whose only routes pass through in-support predecessors (deeper-target case). The mixed configuration, which has no analog in Bloch's framework, is handled in Proposition 2.16 that follows.

Proposition 2.14 (First-target equilibrium)**.** *For a first target i in the support with m_i free-highway incoming edges, and with no in-support predecessor,*

$$x_{i,e} = 1 - \frac{U}{b_i} \quad \textit{on each free-highway edge,} \qquad q_i = \frac{m_i(b_i - U)}{b_i^2}.$$

Proof. Let e be one of the m_i free-highway incoming edges to i. By Lemma 2.12, the attacker's total weight q_i on target i is spread equally across the m_i edges, so the attack probability flowing through edge e is q_i/m_i, and the defender invests the same amount $x_{i,e}$ on each.

Step 1 — defender's first-order condition. Defender i chooses $x_{i,e}$ to minimize its expected loss on edge e plus the cost of that investment. Because e is a free-highway edge, the attacker reaching i through e has met no other defense, so the survival probability of an attack arriving along e is just $(1 - x_{i,e})$. The expected loss from attacks arriving through e is the attack probability through e, times this survival probability, times the value:

$$\text{loss on } e = \frac{q_i}{m_i} \cdot (1 - x_{i,e}) \cdot b_i.$$

The cost of the investment on e is $x_{i,e}^2/2$. Defender i therefore minimizes

$$\frac{q_i}{m_i} \cdot (1 - x_{i,e}) \cdot b_i \; + \; \frac{x_{i,e}^2}{2}.$$

Differentiating with respect to $x_{i,e}$ and setting the derivative to zero:

$$-\frac{q_i}{m_i}\, b_i + x_{i,e} = 0 \quad \Longrightarrow \quad x_{i,e} = \frac{q_i}{m_i}\, b_i. \tag{$*$}$$

Step 2 — attacker's indifference condition. By Lemma 2.6, the attacker's payoff from reaching i through edge e equals the common equilibrium value U. The attacker's payoff along this route is the survival probability times the value, $b_i(1 - x_{i,e})$, so

$$b_i(1 - x_{i,e}) = U \quad \Longrightarrow \quad x_{i,e} = 1 - \frac{U}{b_i}. \tag{$**$}$$

Step 3 — combining. Equation $(**)$ gives the per-edge investment directly. Substituting it into $(*)$ to solve for the attack probability:

$$\frac{q_i}{m_i}\, b_i = 1 - \frac{U}{b_i} \quad \Longrightarrow \quad q_i = \frac{m_i}{b_i}\left(1 - \frac{U}{b_i}\right) = \frac{m_i(b_i - U)}{b_i^2}.$$

Step 4 — feasibility. For q_i to be positive, we need $U < b_i$. If instead $U \geq b_i$, then q_i would be non-positive and i could not be in the support; so the feasibility condition $U/b_i < 1$ holds at every supported first target, and the investment $x_{i,e} = 1 - U/b_i$ is strictly positive there. □

Proposition 2.15 (Deeper-target equilibrium)**.** *For a target i whose only incoming routes in equilibrium pass through in-support predecessors, with the immediate in-support predecessor on each route being a single support member k (by Lemma 2.7, the lowest-value support member from which i can be reached on that route, possibly with non-support shield nodes between k and i) and connecting edge e_i being the edge from k to i along that route,*

$$x_{i,e_i} = 1 - \frac{b_k}{b_i}, \qquad q_i = \frac{b_k(b_i - b_k)}{U \cdot b_i^2}.$$

Proof. Step 1 — attacker's indifference condition. The attacker is indifferent between stopping its attack at k and continuing along the same path to i. Let α_k be the survival probability of reaching k. Attacking k yields a payoff of $\alpha_k b_k$. Continuing to i requires surviving one more edge, e_i, which happens with probability $(1 - x_{i,e_i})$, so attacking i yields $\alpha_k(1 - x_{i,e_i})b_i$. Indifference between the two gives

$$\alpha_k b_k = \alpha_k(1 - x_{i,e_i})b_i \quad \Longrightarrow \quad 1 - x_{i,e_i} = \frac{b_k}{b_i} \quad \Longrightarrow \quad x_{i,e_i} = 1 - \frac{b_k}{b_i}. \tag{$*$}$$

Step 2 — defender's first-order condition. Defender i minimizes its expected loss on edge e_i plus the cost. An attack targeting i arrives along e_i with probability q_i, having already survived up to k with probability α_k; the expected loss is $\alpha_k q_i (1 - x_{i,e_i}) b_i$ and the cost is $x_{i,e_i}^2/2$. Differentiating the sum with respect to x_{i,e_i} and setting it to zero gives the first-order condition

$$x_{i,e_i} = \alpha_k\, q_i\, b_i. \tag{$**$}$$

This has the same structure as the first-target first-order condition, with the arrival probability α_k in place of 1.

Step 3 — combining. From the indifference condition of Step 1, the attacker's payoff from attacking k is $\alpha_k b_k = U$, so $\alpha_k = U/b_k$. Substituting this and equation $(*)$ into $(**)$:

$$1 - \frac{b_k}{b_i} = \frac{U}{b_k}\, q_i\, b_i \quad \Longrightarrow \quad q_i = \frac{b_k}{U b_i}\left(1 - \frac{b_k}{b_i}\right) = \frac{b_k(b_i - b_k)}{U\, b_i^2}.$$

Step 4 — feasibility. For q_i to be positive we need $b_k < b_i$, which is exactly the requirement of Lemma 2.8 that values increase along attack paths. □

We now state the equilibrium equations for the mixed configuration.

Proposition 2.16 (General equilibrium per incoming edge)**.** *Let i be a target in the support, and let $E^{\mathit{free}}(i)$ be its set of free-highway incoming edges and $K(i)$ the set of in-support predecessors from which i is reachable on a route not crossing any other support member. Then in equilibrium:*

- *On each edge $e \in E^{\mathit{free}}(i)$, the investment is $x_{i,e} = 1 - U/b_i$, and the attack probability through e is $(b_i - U)/b_i^2$.*
- *On the edge $e_{k,i}$ connecting predecessor $k \in K(i)$ to i, the investment is $x_{i,e_{k,i}} = 1 - b_k/b_i$, and the attack probability through $e_{k,i}$ is $b_k(b_i - b_k)/(U b_i^2)$.*
- *The total attack probability on i is the sum of the per-edge contributions:*

$$q_i = \frac{m_i(b_i - U)}{b_i^2} + \sum_{k \in K(i)} \frac{b_k(b_i - b_k)}{U\, b_i^2}, \quad \text{where } m_i = |E^{\mathit{free}}(i)|.$$

The first-target case (Proposition 2.14) is the special case $K(i) = \emptyset$; the deeper-target case (Proposition 2.15) is the special case $E^{\mathit{free}}(i) = \emptyset$ and $|K(i)| = 1$.

Proof. Each incoming edge to i in equilibrium is analyzed independently. The two earlier propositions established what the investment and per-edge contribution are for each type of incoming edge: free-highway edges follow the first-target derivation, predecessor edges follow the deeper-target derivation. The arguments are unchanged because the defender's first-order condition on a given edge depends only on the attack probability flowing through that edge and the survival of attacks arriving along it — both of which are determined by the edge's type, not by what happens on other incoming edges of i. Summing the per-edge attack probabilities gives the total q_i. The two earlier propositions correspond to the cases where only one type of incoming edge is present. □

The deeper-target expressions are algebraically identical in form to Bloch's, because a deeper-target predecessor edge is reached through a single connecting edge from its predecessor. The first-target per-edge investment $x_{i,e} = 1 - U/b_i$ also has the same algebraic form as Bloch's first-target investment. But the U inside both expressions is the equilibrium value of the edge-based game, which is not the same number as Bloch's equilibrium value: as the next subsection shows, moving defense to edges generally raises U. The forms are inherited; the magnitudes are not.

Multi-Path Targets Are Structurally Weaker

Holding the equilibrium value U fixed for the moment, compare two first targets in the edge-based model: one with a single free-highway incoming edge, and one with m_i of them. Both invest $1 - U/b_i$ per edge, by Proposition 2.14. The single-edge target pays a defense cost of $(1 - U/b_i)^2/2$; the multi-path target pays

$$C_i = m_i \cdot \frac{(1 - U/b_i)^2}{2},$$

which is m_i times as much. This comparison is a *within-model* statement at a fixed U: it says that, among targets facing the same equilibrium value, the one with more incoming edges carries a proportionally larger defense burden. It is not a comparison with Bloch's equilibrium cost, because Bloch's equilibrium has a different U.

The genuine cross-model effect is precisely that U is not fixed. A multi-path target is cheaper for the attacker to extract value from — the same per-edge defense, spread over more edges, leaves more total attack probability available — and in equilibrium this pushes the attacker's payoff U upward relative to the node-based model. The per-edge investment formula then has the same form as Bloch's but a larger U inside it. Multi-path targets are, in this sense, structurally weaker, and the attacker exploits this by concentrating more attack probability on them.

Strategic Substitutes on Shared Edges

When both endpoints of an edge $e = (i, j)$ have incentive to invest — meaning the edge lies on attack paths targeting both nodes — their best responses interact directly.

Proposition 2.17 (Strategic substitutes)**.** *On a shared edge $e = (i, j)$ where both endpoints have positive attack probability flowing through toward their own node, defender i's best response is*

$$x^*_{i,e} = q\, b_i\, (1 - x_{j,e}),$$

where q is the attack probability through e targeting i. The best responses are strategic substitutes: $\partial x^*_{i,e}/\partial x_{j,e} = -q\, b_i < 0$.

Proof. Defender i chooses $x_{i,e}$ to minimize its loss on edge e plus cost. The survival probability across e is $(1 - x_{i,e})(1 - x_{j,e})$, so the loss term is $q\, b_i\, (1 - x_{i,e})(1 - x_{j,e})$ and the cost term is $x^2_{i,e}/2$. Differentiating their sum with respect to $x_{i,e}$ and setting it to zero:

$$-q\, b_i\, (1 - x_{j,e}) + x_{i,e} = 0 \quad \Longrightarrow \quad x^*_{i,e} = q\, b_i\, (1 - x_{j,e}).$$

Differentiating the best response with respect to the neighbor's investment gives $\partial x^*_{i,e}/\partial x_{j,e} = -q\, b_i$, which is strictly negative whenever attacks flow through the edge. $\square$

This is strategic substitutes: when one endpoint invests more, the other's optimal response is to invest less. The mechanism is intuitive — if my neighbor is already catching most attackers on the connection we share, my own marginal investment matters less — and it is entirely absent from Bloch's model, where one node's investment never enters another node's decision.

A natural concern is whether this dynamic could push both endpoints to invest nothing, leaving the shared edge undefended.

Proposition 2.18 (A shared edge cannot be left undefended in equilibrium)**.** *On a shared edge $e = (i, j)$ with positive attack probability flowing through toward each endpoint, the pair $(x_{i,e}, x_{j,e}) = (0, 0)$ is not an equilibrium of the edge.*

Proof. Suppose $x_{j,e} = 0$. By Proposition 2.17, defender i's best response is $x^*_{i,e} = q\, b_i\,(1 - 0) = q\, b_i$, which is strictly positive because attacks flow through the edge toward i. So $(0, 0)$ is not consistent with defender i best-responding, and cannot be an equilibrium of the edge. $\square$

The remaining questions about strategic substitutes — whether the interior equilibrium is unique, whether corner solutions arise, and whether the interior equilibrium is stable under best-response dynamics, meaning whether iterated best responses by the two endpoints converge back to it after a small perturbation to one endpoint's investment — are left open and noted again in the conclusion.

To illustrate why this mechanism rarely shows up on simple networks, consider a line ATK–T_1–T_2–T_3. The edge (T_1, T_2) lies on the attack path to T_2 and to T_3, but not to T_1, because the attacker has already passed T_1 by the time it traverses this edge. Defender T_1 has no incentive to invest on (T_1, T_2), since attacks through this edge do not target T_1. The same is true for every interior edge of a line: only the downstream node has reason to invest. Strategic substitutes therefore do not arise on lines; they appear only in networks where a single edge serves as the attack route for multiple downstream targets.

Braess Paradox Does Not Occur

In the original model, adding a new connection to the network can sometimes harm the attacker, because the additional vulnerability prompts defenders to increase their investments more than enough to offset the new attack route. This is a version of Braess's paradox.

Proposition 2.19 (No Braess paradox)**.** *Adding a free-highway incoming edge to a supported first target i raises m_i by one and strictly increases the attacker's equilibrium payoff U.*

Proof. Fix the support and consider the equilibrium constraint $\sum_{j\in\Delta} q_j(U) = 1$. By Proposition 2.14, the contribution of target i is $q_i = m_i(b_i - U)/b_i^2$, which is increasing in m_i. Replacing m_i by $m_i + 1$ raises the left-hand side above 1 at the old value of U. Since every q_j is strictly decreasing in U, restoring the equality to 1 requires U to rise. Formally, implicit differentiation of the constraint gives

$$\frac{\partial U}{\partial m_i} = \frac{(b_i - U)/b_i^2}{\sum_{j\in\Delta} |\partial q_j/\partial U|} > 0,$$

since the numerator is positive (because $U < b_i$ at any supported target) and the denominator is a sum of positive terms. □

The reason is that each new edge must be defended separately at the same per-edge level, but the attacker now has more attack vectors, and the per-edge defense cannot rise to compensate because it is already pinned down by the attacker's indifference condition. Adding a connection unambiguously helps the attacker; Braess's paradox in its classical form has no room to operate.

Cooperative Symmetry and Cost Savings

Two distinctive results emerge in the cooperative version of the model. The first is symmetry within edges.

Proposition 2.20 (Symmetry within edges)**.** *In any cooperative equilibrium, the two endpoints of an edge $e = (i, j)$ on which both have positive incentive to invest contribute equally: $y_{i,e} = y_{j,e}$.*

Proof. The contribution of edge $e = (i, j)$ to the cooperative loss function consists of survival terms in which $y_{i,e}$ and $y_{j,e}$ enter only through the symmetric product $(1 - y_{i,e})(1 - y_{j,e})$, plus the two cost terms $y_{i,e}^2/2$ and $y_{j,e}^2/2$. Writing S for the sum of the attack-probability-weighted survival factors that multiply this product (a quantity that does not itself involve $y_{i,e}$ or $y_{j,e}$), the first-order condition for $y_{i,e}$ is

$$y_{i,e} = (1 - y_{j,e})\, S,$$

and the first-order condition for $y_{j,e}$ is the same with i and j exchanged: $y_{j,e} = (1-y_{i,e})\, S$. Subtracting the two:

$$y_{i,e} - y_{j,e} = (1 - y_{j,e})S - (1 - y_{i,e})S = S\,(y_{i,e} - y_{j,e}),$$

so $(1 - S)(y_{i,e} - y_{j,e}) = 0$. Generically $S \neq 1$, so $y_{i,e} = y_{j,e}$. □

The two investments enter the cooperative problem in exactly the same way, so the coordinator has no reason to treat them differently, and the unique solution sets them equal.

The second result is cost savings on lines.

Proposition 2.21 (Cost savings under shared support)**.** *On a line network, when the cooperative equilibria under node-based and edge-based defense share the same support, the two formulations achieve the same attacker utility U, but the centralized defender incurs strictly lower total investment cost under edge-based defense.*

Proof. Suppose the node-based and edge-based cooperative equilibria share the same support. At each position on the line, fix the effective interception so that the attacker's survival probability through that position is the same in both formulations. Since the support is the same, the attacker's indifference conditions are then satisfied at the same value U in both, which establishes the equal-utility claim.

It remains to compare costs. At an interior position, let y be the single investment the node-based coordinator uses. To achieve the same effective interception with two endpoints each investing z, edge-based defense needs $(1-z)^2 = 1-y$, so $z = 1-\sqrt{1-y}$. The node-based cost at that position is $y^2/2$; the edge-based cost is $2 \cdot z^2/2 = z^2$. Writing $u = 1-y \in (0,1)$, so that $y = 1-u$ and $z = 1-\sqrt{u}$, the cost difference is

$$\frac{y^2}{2} - z^2 = \frac{(1-u)^2}{2} - (1-\sqrt{u})^2,$$

which is strictly positive for all $u \in (0,1)$. The entrance edge has only one investor in both formulations, so its cost is unchanged; the saving comes entirely from the interior positions and is strictly positive. The detailed algebra confirming positivity is given in the Appendix. $\square$

The intuition is the convexity of the cost function. Splitting the work of interception between two payers, each doing less, costs less in total than concentrating it on one — and the cooperative defender exploits this at every interior position on the line. The qualifier "under shared support" matters: the result compares the two formulations only when they attack the same set of targets, and Section 2.6 shows by example that this is the regime in which the comparison is clean.

2.6 Numerical Illustrations

To make the effects concrete, we work through two numerical examples. The first illustrates the conditional cost savings result on a line network — the savings that arise when the support is the same under both defense models. The second illustrates multi-path weakness on a small general network. The detailed step-by-step calculations behind both tables, including the iterative identification of the support and the solution of the equilibrium equations, are collected in the Appendix; here we report the results and explain what they mean.

Example 1: A Line of Three Targets (Cooperative Comparison)

Consider the network ATK–T_1–T_2–T_3 with values $b_1 = 0.40$, $b_2 = 0.50$, $b_3 = 0.70$. Under cooperative defense, the centralized defender finds it optimal to drop T_2 from the attack support and use it as a shield. The result is the same attacker utility under both defense models, $U = 0.24$, but with different investment structures, shown in Table 2.1.

	Node Defense (Bloch)	**Edge-Based**
Attacker utility U	0.2400	0.2400
Support	$\{T_1, T_3\}$	$\{T_1, T_3\}$
r_1 (attack probability on T_1)	0.2313	0.5264
r_3 (attack probability on T_3)	0.7687	0.4736
Investment at position 1	0.4000	0.4000 (entrance edge)
Investment at position 2	0.2441 (one node)	0.1307 (per endpoint)
Investment at position 3	0.2441 (one node)	0.1307 (per endpoint)
Effective interception, position 2	0.2441	$1 - (1 - 0.1307)^2 = 0.2441$
Effective interception, position 3	0.2441	0.2441
Total defense cost	0.1396	0.1142
Cost saving	—	18.2%

Table 2.1: Cooperative comparison on a line network with values $(0.40, 0.50, 0.70)$.

The same level of effective interception is achieved at every position. What changes is how that interception is paid for. Under node defense, a single defender bears the full cost of $0.2441^2/2 = 0.0298$ at position 2, and the same again at position 3. Under edge defense, each interior edge receives investment from two endpoints, each paying $0.1307^2/2 = 0.00854$, so the cost per interior edge is 0.0171 rather than 0.0298. The convexity of the quadratic cost function means that splitting investment across two payers is genuinely cheaper than concentrating it on one. The total saving aggregates to 18.2% across the line.

The attack probabilities differ between the two models because the cooperative defender's first-order conditions on interior edges have a different structure under edge defense — they involve an additional $(1 - x)$ factor from the survival product. But the equilibrium attacker utility is the same in both cases here, because the support coincides under both defense models. This is the cleanest demonstration of the cost-savings result:

the attacker is no better or worse off, but defense is delivered more efficiently.

Example 2: A General Network with a Multi-Path Target (Decentralized Comparison)

Consider a small general network: the attacker ATK connects to T_1 and T_2; T_1 connects to T_3 and T_4; T_2 connects to T_4. Values are $b_1 = 0.20$, $b_2 = 0.25$, $b_3 = 0.50$, $b_4 = 0.70$. Target T_4 has two distinct incoming free-highway edges — one from T_1 and one from T_2 — so $m_4 = 2$. Target T_3 has only one incoming edge, so $m_3 = 1$. Targets T_1 and T_2 are first targets reachable directly from the attacker.

	Node Defense (Bloch)	**Edge-Based**
Attacker utility U	0.4020	0.4773
Change in U	—	+18.7%
Support	$\{T_3, T_4\}$	$\{T_3, T_4\}$
m_3	1	1
m_4	1	2
Attack probability q_3	0.3920	0.0909
Attack probability q_4	0.6081	0.9091
Investment x_3	0.1960	0.0455
Investment x_4	0.4257	0.3181 (per edge)
T_3 defense cost	0.0192	0.0010
T_4 defense cost	0.0906	0.1012

Table 2.2: Decentralized comparison on a general network with values $(0.20, 0.25, 0.50, 0.70)$.

The story behind these numbers matters more than the numbers themselves.

The mechanism begins with T_4. Under node defense, T_4 has one investment that covers all directions of approach, and the attacker faces full interception strength regardless of which path is used. Under edge defense, the same target must defend two separate incoming edges, each receiving its own investment. The per-edge investment formula gives $x_{4,e} = 1 - U/b_4$ on each edge, but the total attack probability now scales with $m_4 = 2$.

This raises the attacker's expected payoff U, because T_4 is structurally weaker. The rise in U propagates through the rest of the network. With U now larger, the equilibrium investments at other targets adjust: T_3 in particular finds that defending itself becomes barely worthwhile, since its value $b_3 = 0.50$ is only slightly above the new $U = 0.4773$. The investment at T_3 collapses from 0.1960 to 0.0455.

The attacker, facing a much weaker T_4 with two routes of approach, concentrates almost all attacks there: q_4 rises from 61% to 91%. T_3 becomes a secondary target. The total cost picture is more nuanced — T_4's cost rises slightly because it is now defending two edges, while T_3's cost collapses because it is barely defending at all.

The logical chain is therefore: more incoming edges to a target $\rightarrow$ each edge structurally weaker $\rightarrow$ higher U $\rightarrow$ lower-value targets pushed toward the margin $\rightarrow$ attacks concentrate on the multi-path target. Each step follows from the previous one, and the whole chain is a direct consequence of moving defense to edges.

2.7 Managing a Shared Edge: Four Possibilities

The move from node-based to edge-based defense opens up a question that does not arise in Bloch's model: how exactly is a shared edge managed by its two endpoints? Two modelling choices have to be made, and each can be made in two ways.

The first choice is the *interception technology* — how the two endpoints' investments combine into a survival probability for the edge. Under *independent* interception, each endpoint attempts to intercept on its own and the attacker survives only if both attempts fail, giving $S(e) = (1-x_{i,e})(1-x_{j,e})$. Under *joint* interception, interception requires both endpoints acting together, giving $S(e) = 1 - x_{i,e}\, x_{j,e}$.

The second choice is the *cost structure*. Under *separable* cost, each endpoint pays for its own investment, $x_{i,e}^2/2$ for defender i. Under *shared* cost, the total investment on the edge enters a single combined cost $C(e) = (x_{i,e}+x_{j,e})^2/2$ that is split equally between the two endpoints, so defender i's share of the cost is $(x_{i,e}+x_{j,e})^2/4$.

Crossing the two choices gives four cases, shown in Table 2.3. The main analysis of this chapter has developed one of them in full — separable cost with independent interception. A priori, none of the four is more natural than the others; each describes a coherent way two authorities might share the defense of a connection. We do not develop full equilibrium results for all four, but it is worth setting out, for each, what the defender's best response on a shared edge looks like and what it implies.

In each case below, consider a shared edge $e = (i, j)$ with attack probability q flowing through it toward target i, and let defender i choose $x_{i,e}$ to minimize its loss on the edge plus its cost. The neighbor's investment is $x_{j,e}$. The object whose curvature we examine is the defender's loss function on the edge; a negative cross-partial of that loss function corresponds to substitutes, a positive one to complements.

Case A: separable cost, independent interception. This is the model developed throughout the chapter. Defender i minimizes $q\, b_i\, (1-x_{i,e})(1-x_{j,e}) + x_{i,e}^2/2$, giving the best response

$$x_{i,e}^* = q\, b_i\, (1 - x_{j,e}).$$

	Independent interception $S(e) = (1 - x_{i,e})(1 - x_{j,e})$	**Joint interception** $S(e) = 1 - x_{i,e}x_{j,e}$
Separable cost $x_{i,e}^2/2$ per defender	Case A (the model of this chapter)	Case C
Shared cost $(x_{i,e} + x_{j,e})^2/4$ per defender	Case B	Case D

Table 2.3: Four ways to model the management of a shared edge. Under shared cost, the total cost on the edge $(x_{i,e} + x_{j,e})^2/2$ is split equally between the two endpoints.

The slope $\partial x_{i,e}^*/\partial x_{j,e} = -q\, b_i$ is negative: the loss function is submodular in the two investments, the best responses are strategic substitutes, and by Proposition 2.18 the all-zero outcome is never an equilibrium of the edge.

Case B: shared cost, independent interception. Defender i minimizes $q\, b_i\, (1 - x_{i,e})(1 - x_{j,e}) + (x_{i,e} + x_{j,e})^2/4$. The first-order condition is

$$-q\, b_i\, (1 - x_{j,e}) + \tfrac{1}{2}(x_{i,e} + x_{j,e}) = 0,$$

giving

$$x_{i,e}^* = 2\, q\, b_i\, (1 - x_{j,e}) - x_{j,e}.$$

The slope is $-2\, q\, b_i - 1$, more steeply negative than in Case A. The neighbor's investment hurts defender i twice over: once through the survival term, where the neighbor catching more attackers lowers i's marginal benefit, and once through the shared cost term, where the neighbor spending more raises i's marginal cost. The substitutes effect is stronger, free-riding pressure is intensified, and for a sufficiently large neighbor investment the interior best response turns negative and defender i is pushed to the corner $x_{i,e} = 0$. But the all-zero outcome is still not an equilibrium of the edge: if the neighbor invests zero, i's best response is $2\, q\, b_i > 0$, as in Case A.

Case C: separable cost, joint interception. Here the interception technology changes. Defender i minimizes $q\, b_i\, (1 - x_{i,e}x_{j,e}) + x_{i,e}^2/2$, giving

$$x_{i,e}^* = q\, b_i\, x_{j,e}.$$

The slope $\partial x_{i,e}^*/\partial x_{j,e} = +q\, b_i$ is now *positive*. The sign has flipped: the loss function is supermodular in the two investments, and the best responses are strategic *complements* rather than substitutes. The reason is structural — under joint interception, if the neighbor invests nothing then the product $x_{i,e}\, x_{j,e}$ is zero no matter what defender i does, so i's own investment intercepts nothing and its best response is also zero. Consequently the all-zero outcome *is* an equilibrium of the edge: with joint interception, mutual non-investment is self-sustaining, because neither endpoint can accomplish anything alone.

Case D: shared cost, joint interception. Combining both alternatives, defender i minimizes $q\, b_i\, (1 - x_{i,e}x_{j,e}) + (x_{i,e} + x_{j,e})^2/4$. The first-order condition is

$$-q\, b_i\, x_{j,e} + \tfrac{1}{2}(x_{i,e} + x_{j,e}) = 0,$$

giving

$$x^*_{i,e} = (2\, q\, b_i - 1)\, x_{j,e}.$$

The slope is $2\, q\, b_i - 1$, and its sign is a genuine threshold: the best responses are complements if $q\, b_i > 1/2$ and substitutes if $q\, b_i < 1/2$. Case D is a competition between two forces — the complementarity coming from joint interception, and the substitutability coming from the shared marginal cost — and which one wins depends on the magnitude of $q\, b_i$. Under the model's inherited normalization $b_i \in (0, 1)$, together with $q \leq 1$, we have $q\, b_i < 1$ but $q\, b_i$ can lie on either side of the $1/2$ threshold: when the attack probability on a high-value target is sufficient (specifically when $q\, b_i > 1/2$), the case sits in the complements regime, and otherwise in the substitutes regime. As in Case C, joint interception makes the all-zero outcome an equilibrium of the edge.

What the four cases show. The clearest message of the table is that the two modelling choices are not equally consequential. Changing the cost structure — from separable to shared — only modulates the *strength* of free-riding within a given interception technology: it makes substitutes stronger, or it modulates the complement/substitute trade-off, but it does not change whether an undefended edge can persist. Changing the interception technology, on the other hand, changes the *shape* of the equilibrium set. Under independent interception (Cases A and B), the all-zero outcome is never an equilibrium of the edge, because a single endpoint investing alone still intercepts something. Under joint interception (Cases C and D), the all-zero outcome becomes an equilibrium, because a single endpoint acting alone accomplishes nothing, and mutual non-investment sustains itself. The probability dimension, in other words, is the structurally decisive one.

Two qualifications should accompany this discussion. First, the all-zero statements above are statements about the *edge-level subgame*: they say a shared edge, considered on its own, admits a zero-investment equilibrium. Whether the full network game has an equilibrium with that edge undefended is a separate question that depends on the attacker's mixing and on whether attacks actually flow through the edge. Second, the threshold structure of Case D — the sign of $2\, q\, b_i - 1$ — means the case genuinely moves between the complement and substitute regimes within the model's own parameter range, making it a bifurcation regime worth study in its own right.

Conclusion to the Chapter

This chapter has built the modified model from the ground up, established its formal properties with proofs given alongside each statement, and illustrated its behavior on two worked examples. The structural changes are concentrated around three places: the survival formula on individual edges, the new factor m_i that captures multi-path weakness at first targets, and the strategic substitutes mechanism on shared edges. From these three changes the rest follows — amplified responses to the network structure, the disappearance of Braess's paradox, the symmetry of cooperative defense within edges, and the conditional cost savings on lines. The closing section showed that the management of a shared edge can be modelled in four ways, and that the choice of interception technology, more than the choice of cost structure, is what determines whether a shared connection can end up undefended in equilibrium. The implementation that supports these results numerically is the subject of Chapter 4, and Chapter 3 extends the analysis to two settings of incomplete information before that.

Chapter 3

Incomplete-Information Extensions

The analysis of the previous chapter rests on full common knowledge. Every defender knows every target's value; every player observes every edge in the network; the attacker chooses a mixed strategy against fully transparent defenders, and each defender best-responds against neighbors whose investments it can compute exactly. The structural results — strategic substitutes on shared edges, mutual coverage of free-highway incoming edges, the trade-off between first-target and deeper-target investments — all rest on this common-knowledge assumption.

This chapter relaxes it in two distinct ways. In the first extension, the network structure is still common knowledge, but defenders' target values become private types drawn from a known prior. In the second, target values are common knowledge, but the network structure is only partially observed by each player. The two cases isolate the role of each kind of uncertainty.

Cybersecurity provides a natural setting for both. The empirical study by Chang et al. (2024) documents that attackers in the United States target firms based on observable signals — size, sales, public-vs-private status — rather than on the actual value at stake within each firm. A firm knows its own data, revenue, and exposure to ransom; the attacker does not. Symmetrically, in a compartmentalized organization or in intelligence operations, each player typically knows the connections it participates in directly but has only a probabilistic view of connections elsewhere. The two extensions in this chapter map onto these two information asymmetries.

The framework throughout is Bayesian: each player has a type (their private information), a strategy that may depend on what they observe, and a best response that takes expectations over what they do not observe. The equilibrium concept is Bayes-Nash, following Vega-Redondo (2003, Ch. 6). The aim is not full analytical generality but to establish that the edge-based defense model retains tractable structure under each kind of uncertainty, and to derive worked examples that illustrate how the central mechanisms change.

A caveat applying to both extensions. The worked examples use the triangle network because it is the smallest network where the shared-edge mechanism appears at all. The triangle's three edges and two defenders make the algebra manageable, but they also leave a great deal of structure on the table. Larger networks add features that simply do not exist in the triangle: longer attack paths, deeper-target routing, multi-path coverage, free-highway edges that pass through several non-support nodes, and combinations of these. The mechanisms identified in this chapter — the moment-based equilibrium dependence in the value case, the threshold structure in the network case, the comparisons with the complete-information baseline — should be read as a first picture rather than the full one. Extending the analysis to richer networks is a natural next step and would likely reveal interactions between incomplete information and the structural results of Section 2.3 that the triangle is too small to expose.

3.1 Private Target Values

Model setup

The network (V, E), the attacker located at node 0, and the cost structure are all retained from the baseline edge-based model. The change is in what each player observes.

Definition 3.1 (Information structure — value case). *Each defender $i \in \{1, \ldots, n\}$ has a private type $b_i \in (0, 1)$, drawn independently from a common prior distribution F with finite mean $\mu = \mathbb{E}_F[b]$ and finite second moment $m = \mathbb{E}_F[b^2]$. The prior F is common knowledge. The realisation b_i is observed only by defender i: defender i observes b_i but not b_j for any $j \neq i$. The attacker observes the network structure and the prior F but not the realised type profile $(b_1, \ldots, b_n)$.*

In the baseline model, the genericity assumption rules out ties among target values; here it is replaced by the prior, which assigns probability zero to ties for any continuous F. So the spirit of the assumption is preserved without needing to invoke it explicitly.

Definition 3.2 (Strategies). *A pure strategy for defender i is a type-contingent investment function*

$$x_i : (0, 1) \to [0, 1]^{|E(i)|}$$

mapping defender i's realised type to a vector of edge investments. A mixed strategy for the attacker is a probability distribution π over paths, chosen without conditioning on the realised type profile.

Definition 3.3 (Bayes-Nash equilibrium — value case). *A profile $(\pi^*, \{x_i^*\}_{i=1}^n)$ is a Bayes-Nash equilibrium if (i) each defender i's strategy $x_i^*(b_i)$ minimises its interim expected loss given its own realised type b_i, taking expectations over neighbors' types using F, and (ii) the attacker's strategy π^* maximises its ex-ante expected payoff, taking expectations over the full type profile.*

Shared-edge best response

The central building block of the model is the defender's best response on a shared edge. Under complete information, this is given by the Case A formula of Section 2.7: defender i on shared edge $e = \{i, j\}$ with attack probability q flowing through it chooses

$$x^*_{i,e} = q \cdot b_i \cdot (1 - x_{j,e}).$$

Defender i's investment depends on the neighbor's actual investment $x_{j,e}$, which in turn depends on the neighbor's value b_j.

Under incomplete information, defender i does not know b_j. It minimises its expected loss, averaging over the prior:

$$\min_{x_{i,e}\in[0,1]} \mathbb{E}_{b_j\sim F}\left[q \cdot b_i \cdot (1 - x_{i,e})(1 - x^*_{j,e}(b_j)) + \frac{x^2_{i,e}}{2}\right].$$

Since only the second survival factor depends on b_j, the expectation factors:

$$q \cdot b_i \cdot (1 - x_{i,e}) \cdot (1 - \bar{x}_j) + \frac{x^2_{i,e}}{2}, \quad \text{where } \bar{x}_j = \mathbb{E}_{b_j\sim F}[x^*_{j,e}(b_j)].$$

Differentiating and setting to zero gives the type-contingent best response

$$x^*_{i,e}(b_i) = q \cdot b_i \cdot (1 - \bar{x}_j).$$

The only change from the complete-information formula is that the neighbor's actual investment $x_{j,e}$ has been replaced by its expected value $\bar{x}_j$. Defender i reacts to the neighbor's expected behaviour rather than to its actual behaviour. This is the standard observation in Bayesian games (Vega-Redondo, 2003, Ch. 6): each type of each player acts optimally against the type-averaged behaviour of the others.

Worked example: the triangle network

To make the mechanism concrete, consider the triangle network: nodes ATK, T_1, T_2 with edges $\{\mathrm{ATK}, T_1\}$, $\{\mathrm{ATK}, T_2\}$, and the shared edge $\{T_1, T_2\}$.

The attacker has four routes: direct to T_1, direct to T_2, roundabout to T_1 via T_2, and roundabout to T_2 via T_1. Denote the corresponding mixed-strategy probabilities $\alpha_1, \alpha_2, \gamma_1, \gamma_2$, with

$$\alpha_1 + \alpha_2 + \gamma_1 + \gamma_2 = 1. \tag{3.1}$$

Under the symmetric prior, the equilibrium is symmetric: $\alpha_1 = \alpha_2 = \alpha$ and $\gamma_1 = \gamma_2 = \gamma$. Equation (3.1) then reduces to

$$2\alpha + 2\gamma = 1. \tag{3.2}$$

Direct edge. Only the defender invests on the direct edge from ATK; the attacker does not. The defender's expected loss on this edge plus the cost is

$$\alpha \cdot b \cdot (1 - a(b)) + \frac{a(b)^2}{2}. \tag{3.3}$$

Differentiating Equation (3.3) with respect to $a(b)$ and setting the derivative to zero gives the first-order condition $-\alpha \cdot b + a(b) = 0$, so

$$a(b) = \alpha \cdot b. \tag{3.4}$$

Taking the expectation of Equation (3.4) over $b \sim F$ gives the mean direct-edge investment:

$$\bar{a} = \mathbb{E}[a(b)] = \alpha \mathbb{E}[b] = \alpha\mu. \tag{3.5}$$

Shared edge. Both endpoints invest on the shared edge. Defender 1, with value b_1, faces attacks targeting T_1 along the roundabout route ATK $\rightarrow T_2 \rightarrow T_1$, with probability γ. Survival across this route is $(1 - a(b_2)) \cdot (1 - s(b_1)) \cdot (1 - s(b_2))$. Defender 1 does not observe b_2, so it averages over $b_2 \sim F$. Defender 1's expected loss from this edge plus the cost is

$$\gamma \cdot b_1 \cdot \mathbb{E}_{b_2}[(1 - a(b_2)) \cdot (1 - s(b_1)) \cdot (1 - s(b_2))] + \frac{s(b_1)^2}{2}. \tag{3.6}$$

Since $(1 - s(b_1))$ does not depend on b_2, it factors out of the expectation:

$$\gamma \cdot b_1 \cdot (1 - s(b_1)) \cdot C + \frac{s(b_1)^2}{2}, \quad \text{where } C \equiv \mathbb{E}_b[(1 - a(b))(1 - s(b))]. \tag{3.7}$$

The factor C captures the joint defense the other endpoint provides on average, including both its direct-edge investment (which protects approach to itself) and its shared-edge investment.

Differentiating Equation (3.7) with respect to $s(b_1)$ and setting the derivative to zero gives the first-order condition $-\gamma \cdot b_1 \cdot C + s(b_1) = 0$, so

$$s(b) = \gamma \cdot b \cdot C. \tag{3.8}$$

Taking the expectation of Equation (3.8):

$$\bar{s} = \mathbb{E}[s(b)] = \gamma\mu C. \tag{3.9}$$

Computing C

Expanding the definition of C from Equation (3.7):

$$
\begin{aligned}
C &= \mathbb{E}[(1-a(b))(1-s(b))] \\
&= 1 - \bar{a} - \bar{s} + \mathbb{E}[a(b)s(b)]. \qquad (3.10)
\end{aligned}
$$

The first three terms are direct from Equations (3.5) and (3.9). For the cross term, substitute Equations (3.4) and (3.8):

$$\mathbb{E}[a(b)s(b)] = \mathbb{E}[(\alpha b)(\gamma bC)] = \alpha\gamma C\mathbb{E}[b^2] = \alpha\gamma C m. \qquad (3.11)$$

Substituting Equations (3.5), (3.9), and (3.11) into Equation (3.10):

$$C = 1 - \alpha\mu - \gamma\mu C + \alpha\gamma C\, m. \qquad (3.12)$$

Solving Equation (3.12) for C:

$$C = \frac{1-\alpha\mu}{1+\gamma\mu-\alpha\gamma m}. \qquad (3.13)$$

Attacker's indifference conditions

Both path types (direct and roundabout) must yield the same expected payoff U in equilibrium, otherwise the attacker would shift mass toward the better path. The direct-route expected payoff to T_1 is the expectation of the attacker's payoff $b_1 \cdot (1-a(b_1))$ over $b_1 \sim F$:

$$\mathbb{E}[b(1-a(b))] = \mathbb{E}[b] - \mathbb{E}[b \cdot a(b)]. \qquad (3.14)$$

Substituting Equation (3.4) into Equation (3.14):

$$\mathbb{E}[b(1-a(b))] = \mu - \mathbb{E}[\alpha b^2] = \mu - \alpha m. \qquad (3.15)$$

So the equilibrium condition on the direct route is

$$U = \mu - \alpha m. \qquad (3.16)$$

For the roundabout to T_1, the attacker's payoff is $b_1 \cdot (1-a(b_2)) \cdot (1-s(b_1)) \cdot (1-s(b_2))$. Since b_1 and b_2 are independent draws from F, the expectation factors:

$$\mathbb{E}_{b_1,b_2}[b_1(1-a(b_2))(1-s(b_1))(1-s(b_2))] = \mathbb{E}_{b_1}[b_1(1-s(b_1))] \cdot \mathbb{E}_{b_2}[(1-a(b_2))(1-s(b_2))]. \qquad (3.17)$$

The second factor on the right of Equation (3.17) is exactly C, by Equation (3.7). The first factor evaluates as follows: substituting Equation (3.8),

$$\mathbb{E}_{b_1}[b_1(1-s(b_1))] = \mathbb{E}[b] - \mathbb{E}[b \cdot s(b)] = \mu - \gamma C m. \tag{3.18}$$

Combining Equations (3.17) and (3.18):

$$U = (\mu - \gamma C m) \cdot C. \tag{3.19}$$

Equations (3.16), (3.19), (3.2), and (3.13) form a closed system in the four unknowns (α, γ, C, U).

Numerical solution

For concreteness, take F to be uniform on $(0.2, 0.8)$. Then $\mu = 0.5$ and $m = 0.28$. Solving the system numerically: The type-contingent rules are $a(b) \approx 0.455\, b$ and $s(b) \approx 0.034\, b$.

α (direct attack probability per target)	0.4550
γ (roundabout attack probability per direction)	0.0450
C (joint-defense factor)	0.7598
U (attacker payoff)	0.3726
$\bar{a}$ (mean direct-edge investment)	0.2275
$\bar{s}$ (mean shared-edge investment)	0.0171

Table 3.1: Equilibrium values for the value-incomplete triangle (uniform prior on $(0.2, 0.8)$).

Comparison with the complete-information baseline

For a direct comparison, consider the complete-information triangle with $b_1 = 0.49$ and $b_2 = 0.51$. These values satisfy the genericity assumption ($b_1 \neq b_2$) and sit symmetrically around the mean of the prior used above. Both targets are in the support. By Lemma 2.8 (values increase along attack paths), the predecessor relationships are: T_2 is reached both via the direct edge from ATK (free-highway) and via the predecessor edge from T_1, while T_1 is reached only via its direct edge from ATK (the predecessor edge from T_2 would violate values-increase, since $b_2 > b_1$). So T_2 is in the mixed configuration handled by Proposition 2.16; T_1 is a pure first target with $m_1 = 1$.

Solving the complete-information system gives the values in the second column below. The third column repeats the incomplete-information equilibrium under the uniform prior on $(0.2, 0.8)$.

Three observations stand out.

	Complete info ($b_1 = 0.49, b_2 = 0.51$)	Incomplete info ($\mu = 0.5$)
U (attacker payoff)	0.3869	0.3726
x on $\{\text{ATK}, T_1\}$	0.2104	$a(0.49) = 0.2229$
x on $\{\text{ATK}, T_2\}$	0.2414	$a(0.51) = 0.2320$
x on $\{T_1, T_2\}$ by T_1	0.0000	$s(0.49) = 0.0168$
x on $\{T_1, T_2\}$ by T_2	0.0392	$s(0.51) = 0.0175$
Total attack on shared edge	$\approx 9.7\%$ (via predecessor)	$\approx 9.0\%$ (via roundabout)

Table 3.2: Complete- vs incomplete-information equilibrium on the triangle, $b_1 = 0.49$, $b_2 = 0.51$.

The shared edge is defended in both cases, but for different reasons. Under complete information, the shared edge is defended by T_2 alone, because T_1 is the lower-value target and has no incentive to invest on an edge whose attacks target only T_2. Under incomplete information, both endpoints invest on the shared edge, because neither knows with certainty that it is the lower-value target.

The investment pattern is more symmetric under incomplete information. The complete-information equilibrium has a sharp asymmetry: T_2 invests 0.039 on the shared edge while T_1 invests zero. The incomplete-information equilibrium has both defenders investing nearly the same amount on the shared edge (both close to 0.017), because each defender averages over the possibility of being the higher-value or lower-value type.

The attacker is worse off under incomplete information. The attacker's expected payoff falls from 0.3869 to 0.3726 when moving from complete to incomplete information. This effect runs opposite to a common intuition that "the attacker exploits the defenders' uncertainty." Here, the symmetry-breaking under complete information — where one defender bears the full burden of shared-edge defense — gives the attacker more leverage than the incomplete-information case, where uncertainty forces both defenders to contribute and the joint coverage of the shared edge is slightly stronger overall.

What carries over and what changes

The shared-edge result above is the simplest illustration of a general pattern. More broadly:

Within a single target, mutual coverage plausibly survives. Modified Lemma 2.12 says that a first target with multiple free-highway incoming edges defends each equally, since the attacker would otherwise exploit any imbalance. The intuition suggests this carries over to the within-target level under incomplete information: an individual defender, given its own realised value, would have no reason to favour one of its identical incoming edges over another. However, a formal statement requires a proof tailored to the Bayes-Nash setting, where symmetry arguments become more delicate. We state this as a conjecture rather than a result.

Across targets, indifference becomes expectational. The attacker's mixed strategy no longer equates pointwise payoffs across targets; it equates expected payoffs, with the expectation taken over the prior. Equilibrium quantities depend on distribution-level statistics (μ, m) rather than on individual realisations.

Equilibrium quantities are functions of moments of the prior in symmetric settings. The first-target equation,

$$q_i = \frac{m_i(b_i - U)}{b_i^2},$$

contains b_i explicitly. In symmetric settings of the kind analysed above, where defenders are structurally identical and a linear investment rule $x(b) \propto b$ arises naturally, the analog under incomplete information replaces b_i with moments of the prior:

$$q_i = \frac{m_i^2(\mu - U)}{m}, \qquad \text{where } m = \mathbb{E}[b^2].$$

This expression should be read as the analog in a symmetric environment, not as a general theorem: for heterogeneous priors or asymmetric network positions, attack probabilities will generally depend on more than the first two moments of the prior. Establishing the general statement requires the broader Bayes-Nash framework developed in future work.

Defenders' investments remain type-contingent. Each defender knows its own value and tailors its investment accordingly. The function $x^*(b)$ is monotonic in b in the cases analysed so far — higher-value defenders invest more — which preserves the qualitative comparative statics of the baseline model.

3.2 Partial Network Observation

Model setup

The network (V, E) refers to the actual network in the world: in the worked example below, the full triangle on nodes $\{0, T_1, T_2\}$. Target values are common knowledge. The cost structure is retained from the baseline edge-based model.

Definition 3.4 (Network observation structure)**.** *For each player p, let $V(p) \subseteq E$ denote the set of edges directly incident to p's position in the network. Player p observes $V(p)$*

with certainty: each edge in $V(p)$ is known to exist. For each edge $e \notin V(p)$, player p assigns prior probability $\theta \in (0,1)$ to the event that e exists, with θ common knowledge. Edge realizations are independent across edges under this prior.

Three observations about this setup:

First, the actual network is fixed: every edge in E exists in the realized world. The probability θ describes a player's subjective belief about edges it cannot see, not a genuine randomness in the network. From an outside observer's perspective, the equilibrium will be played on the full network; from each player's perspective, it is played against partial information about which edges are available.

Second, θ is the same for every player and every unseen edge. This is the simplest specification — a single parameter governing network uncertainty. Richer specifications (player-specific priors, correlated edges, edges with different prior probabilities) are natural extensions and are mentioned at the end of this chapter.

Third, target values are common knowledge in this extension. The combined case where both values and network structure are private is a separate problem.

Definition 3.5 (Strategies — network case)**.** *A pure strategy for defender i is a vector of investments on the edges i observes, $x_i \in [0,1]^{|V(i)|}$. A defender's strategy is not type-contingent in this extension: each defender always observes the same edges (its own), so there is one "type" of each defender. A mixed strategy for the attacker is a probability distribution π over the paths it considers possible. The attacker's strategy may place positive probability on paths that traverse edges it does not directly observe, with the understanding that such paths succeed only if those edges actually exist.*

Definition 3.6 (Bayes-Nash equilibrium — network case)**.** *A profile $(\pi^*, \{x_i^*\})$ is a Bayes-Nash equilibrium if (i) each defender i's strategy x_i^* minimizes its expected loss given its observation $V(i)$ and the prior θ over edges in $E \setminus V(i)$, anticipating the equilibrium strategies of the other players, and (ii) the attacker's strategy π^* maximizes its expected payoff given its observation $V(ATK)$ and the prior θ, anticipating the equilibrium strategies of the defenders.*

In the baseline model with complete information, no ties between target values are needed because the game is solved for a specific value vector. Here too, ties are not an issue: target values are fixed and common knowledge, and the analysis proceeds for that given vector. The role of the genericity assumption from the baseline — ruling out knife-edge cases in equilibrium computation — carries over largely unchanged, though the presence of θ introduces an additional smoothing mechanism on top of it.

Worked example: the triangle network

The actual network is the triangle on nodes $\{0, T_1, T_2\}$ with edges $\{0, T_1\}$, $\{0, T_2\}$, and $\{T_1, T_2\}$. Target values are $b_1 = 0.49$ and $b_2 = 0.51$.

Observation structure. Defender 1 (at node T_1) observes the edges incident to T_1: $\{0, T_1\}$ and $\{T_1, T_2\}$. Defender 1's unseen edge is $\{0, T_2\}$, which it believes exists with probability θ.

Defender 2 (at node T_2) observes the edges incident to T_2: $\{0, T_2\}$ and $\{T_1, T_2\}$. Defender 2's unseen edge is $\{0, T_1\}$, with belief θ.

The attacker (at node 0) observes the edges incident to 0: $\{0, T_1\}$ and $\{0, T_2\}$. The attacker's unseen edge is $\{T_1, T_2\}$, with belief θ.

Attacker's strategy space. The attacker considers four possible paths to attack:

- *Direct to T_1:* path 0–T_1, using edge $\{0, T_1\}$. Always available (attacker observes this edge).
- *Direct to T_2:* path 0–T_2, using edge $\{0, T_2\}$. Always available.
- *Roundabout to T_1:* path 0–T_2–T_1, using edges $\{0, T_2\}$ and $\{T_1, T_2\}$. The attacker is uncertain about $\{T_1, T_2\}$, so this path succeeds (in the attacker's expectation) only with probability θ.
- *Roundabout to T_2:* path 0–T_1–T_2, using $\{0, T_1\}$ and $\{T_1, T_2\}$. Also conditional on $\{T_1, T_2\}$ existing.

Denote the attacker's mixed-strategy probabilities by $\alpha_1, \alpha_2, \gamma_1, \gamma_2$, with

$$\alpha_1 + \alpha_2 + \gamma_1 + \gamma_2 = 1. \tag{3.20}$$

Defenders' investment problem. Each defender invests on the two edges it observes. Defender 1 chooses (x_1^{01}, x_1^{12}); defender 2 chooses (x_2^{02}, x_2^{12}).

Crucially, each defender bears losses only when attacks reach *its own target.* An attack on edge $\{0, T_1\}$ may belong to one of two flows: a direct attack on T_1 (with probability α_1, targeting defender 1), or a roundabout attack on T_2 (with probability γ_2, targeting defender 2). Defender 1's investment on $\{0, T_1\}$ affects survival of both flows, but only the first enters defender 1's loss function: losses suffered by T_2 do not appear in T_1's payoff. This separation is what produces the strategic-substitutes mechanism in Section 2.3 and continues to operate here.

Defender 1's direct edge. Defender 1's expected loss on edge $\{0, T_1\}$ comes only from the α_1 traffic:

$$\alpha_1 \cdot b_1 \cdot (1 - x_1^{01}) + \frac{(x_1^{01})^2}{2}. \tag{3.21}$$

Differentiating Equation (3.21) with respect to x_1^{01} and setting it to zero gives

$$x_1^{01} = \alpha_1 \cdot b_1. \tag{3.22}$$

Defender 1's shared edge. The shared edge $\{T_1, T_2\}$ carries two flows: roundabout attacks on T_1 via T_2 (with probability γ_1), and roundabout attacks on T_2 via T_1 (with probability γ_2). Defender 1 cares only about the first.

The γ_1 flow's survival to T_1, conditional on the edge $\{0, T_2\}$ actually existing, is $(1 - x_2^{02})(1 - x_1^{12})(1 - x_2^{12})$. Defender 1 does not observe $\{0, T_2\}$ and assigns probability θ to its existence. So defender 1's expected loss from the shared edge is

$$\gamma_1 \cdot b_1 \cdot \theta \cdot (1 - x_2^{02})(1 - x_1^{12})(1 - x_2^{12}) + \frac{(x_1^{12})^2}{2}. \tag{3.23}$$

The $(1 - x_2^{02})$ and $(1 - x_2^{12})$ terms are anticipated by defender 1 at their equilibrium values — defender 1 does not observe them directly but correctly anticipates them in equilibrium, as is standard in Bayes-Nash.

Differentiating Equation (3.23) with respect to x_1^{12} and setting it to zero:

$$x_1^{12} = \gamma_1 \cdot b_1 \cdot \theta \cdot (1 - x_2^{02})(1 - x_2^{12}). \tag{3.24}$$

Defender 2. By symmetric reasoning,

$$x_2^{02} = \alpha_2 \cdot b_2, \tag{3.25}$$

$$x_2^{12} = \gamma_2 \cdot b_2 \cdot \theta \cdot (1 - x_1^{01})(1 - x_1^{12}). \tag{3.26}$$

Attacker's indifference conditions

The attacker's expected payoff along each path depends on whether the path requires an unseen edge. For paths using only observed edges (direct routes), the attacker computes its payoff deterministically. For paths requiring the unseen edge $\{T_1, T_2\}$ (roundabout routes), the attacker scales its payoff by θ — its belief that the path is actually feasible. This simple θ-discounting structure is specific to the triangle, where each roundabout path requires exactly one unseen edge. In larger networks with multiple unseen edges per path, the discount would involve products of probabilities (or, if existence is correlated, conditional probabilities), and the equilibrium analysis would change accordingly.

$$U_{d1} = b_1(1 - x_1^{01}), \tag{3.27}$$
$$U_{d2} = b_2(1 - x_2^{02}), \tag{3.28}$$
$$U_{r1} = \theta \cdot b_1(1 - x_2^{02})(1 - x_1^{12})(1 - x_2^{12}), \tag{3.29}$$
$$U_{r2} = \theta \cdot b_2(1 - x_1^{01})(1 - x_1^{12})(1 - x_2^{12}). \tag{3.30}$$

In equilibrium, every path the attacker uses with positive probability must yield the same expected payoff U. Paths not used must yield expected payoff no greater than U.

Solving the equilibrium

The system has seven unknowns: $\alpha_1, \alpha_2, \gamma_1, \gamma_2$, x_1^{12}, x_2^{12}, and U. (The direct-edge investments x_1^{01}, x_2^{02} are determined by Equations (3.22) and (3.25) once the alpha's are known.)

A clean structural result emerges from looking at when the roundabout routes are profitable for the attacker. Suppose the attacker uses only direct routes ($\gamma_1 = \gamma_2 = 0$). Then no shared-edge attacks reach either target, and the defenders rationally set $x_1^{12} = x_2^{12} = 0$. The remaining four-unknown system is the standard two-direct-route problem:

$$b_1(1 - x_1^{01}) = b_2(1 - x_2^{02}) = U, \quad \alpha_1 + \alpha_2 = 1. \tag{3.31}$$

Solving the direct-only system together with the FOCs of Equations (3.22) and (3.25) gives

$$\alpha_1 = \frac{b_1 - b_2 + b_2^2}{b_1^2 + b_2^2}, \quad \alpha_2 = 1 - \alpha_1. \tag{3.32}$$

When is this direct-only corner the equilibrium? The corner is a Bayes-Nash equilibrium whenever the attacker has no profitable deviation to a roundabout route — a necessary condition for the corner to survive, though we do not establish uniqueness here. Since the defenders are not investing on the shared edge in the corner, the roundabout payoffs become

$$U_{r1} = \theta \cdot b_1(1 - x_2^{02}), \quad U_{r2} = \theta \cdot b_2(1 - x_1^{01}). \tag{3.33}$$

The corner survives if $U_{r1} \le U$ and $U_{r2} \le U$. This gives two thresholds:

$$\theta_1^* = \frac{U}{b_1(1 - x_2^{02})}, \quad \theta_2^* = \frac{U}{b_2(1 - x_1^{01})}. \tag{3.34}$$

The corner survives as an equilibrium for all $\theta \le \min(\theta_1^*, \theta_2^*)$. For larger θ, the attacker would deviate to the more attractive roundabout route, and the equilibrium shifts to a partial-corner regime where one of the gammas turns positive.

Numerical results

For $b_1 = 0.49, b_2 = 0.51$, the corner equilibrium has $\alpha_1 = 0.4800$, $\alpha_2 = 0.5200$, $U = 0.3748$, $x_1^{01} = 0.2352$, $x_2^{02} = 0.2652$. The two thresholds work out to $\theta_1^* = 1.041$ and $\theta_2^* = 0.961$. Since $\theta_1^* > 1$, the roundabout to T_1 is never profitable in the relevant range; the active threshold is $\theta_2^* = 0.961$.

Regime	Range of θ
Direct-only equilibrium ($\gamma_1 = \gamma_2 = 0$)	$\theta \in (0, 0.961)$
Partial roundabout ($\gamma_1 = 0$, $\gamma_2 > 0$)	$\theta \in [0.961, 1]$

Table 3.3: Equilibrium regimes for the network-incomplete triangle, $b_1 = 0.49$, $b_2 = 0.51$.

In the partial-roundabout regime, defender 1 still invests zero on the shared edge ($x_1^{12} = 0$, since $\gamma_1 = 0$ means no attacks come to T_1 via that route). Only defender 2 invests on the shared edge, since it is the only one whose target is reached through it in equilibrium.

Solving the partial-corner system at several values of θ:

θ	α_1	α_2	γ_2	x_2^{12}	U
0.961 (threshold)	0.4800	0.5200	0.0000	0.0000	0.3748
0.980	0.4538	0.4958	0.0504	0.0196	0.3810
0.990	0.4412	0.4842	0.0746	0.0295	0.3841
1.000	0.4294	0.4732	0.0974	0.0392	0.3869

Table 3.4: Partial-corner equilibrium of the network-incomplete triangle at several values of θ.

The $\theta = 1$ row is computed by solving the partial-corner system at $\theta = 1$ — not assumed to match the complete-information case. The fact that it does match (the same $U = 0.3869$ and the same investments derived for the complete-information triangle in Section 3.1) is a consistency check on the construction: at full certainty, the attacker is indifferent between believing the unseen edge exists and observing that it exists, and the equilibria coincide.

What carries over and what changes

Defender FOCs retain their structure in the triangle, with θ entering only on the shared edge. The direct-edge first-order condition is $x_i^{0i} = \alpha_i b_i$, identical in form to the complete-information baseline once α_i is interpreted as that defender's direct-attack probability. The shared-edge FOC carries an extra θ factor because the roundabout attack's feasibility is itself uncertain from the defender's perspective. In larger networks, the FOCs would gain additional θ factors wherever an attack route includes more than one edge the defender cannot see.

The attacker's indifference becomes a comparison between deterministic and θ-discounted payoffs. In the triangle, paths using only observed edges enter the attacker's calculation deterministically, while paths requiring an unseen edge are scaled by θ. This is a per-path adjustment, not a global moment-based one as in the value case. Generalising this beyond the triangle is more involved because a path may rely on several unseen edges with potentially correlated existence.

A new feature in this geometry: a threshold below which uncertainty becomes "absorbing." The value-incomplete case produced a smooth equilibrium dependence on the prior's first two moments. The network case has a sharp threshold below which the equilibrium becomes constant, at least in the triangle studied here. Whether this absorption persists, sharpens, or fragments into multiple thresholds in larger networks is an open question.

3.3 Comparison and Discussion

Placing the two extensions side by side with the complete-information baseline on the same triangle:

Information regime	U
Complete information ($b_1 = 0.49, b_2 = 0.51$)	0.3869
Value-incomplete information (uniform prior, $\mu = 0.5$)	0.3726
Network-incomplete information ($\theta = 1$)	0.3869
Network-incomplete information ($\theta = 0.5$)	0.3748
Network-incomplete information ($\theta = 0.1$)	0.3748

Table 3.5: Attacker payoff U under the three information regimes, on the triangle with $b_1 = 0.49$ and $b_2 = 0.51$.

Three observations stand out from the comparison.

Both kinds of uncertainty hurt the attacker, but through different mechanisms. Under value uncertainty, the attacker's payoff falls because defenders' uncertainty about each other spreads shared-edge defense across both endpoints rather than concentrating it on the higher-value defender. Coverage becomes more uniform, and the joint defense improves slightly. Under network uncertainty, the attacker's payoff falls because roundabout routes drop out of the attacker's effective strategy space when θ falls below the threshold — the attacker simply cannot rely on the shared edge being available, so it does not try to use it.

Value uncertainty hurts the attacker more than network uncertainty, in this triangle. The value-incomplete case at $\mu = 0.5$ gives $U = 0.3726$; the network case at $\theta = 0.5$ gives $U = 0.3748$. The difference is small but interpretable. Under value uncertainty, the defenders actively contribute additional defense (the symmetric shared-edge investment of about 0.017 on each side). Under network uncertainty in the absorbing regime, the shared edge is left undefended, and the attacker's payoff loss comes only from not using the roundabout.

Network uncertainty has a sharp threshold; value uncertainty has smooth dependence on the prior. The value-incomplete equilibrium depends smoothly on the prior's first two moments (μ, m). The network-incomplete equilibrium is piecewise constant: equal to the corner value $U = 0.3748$ for all $\theta < 0.961$, and rising smoothly to 0.3869 above the threshold. This is a qualitative difference between the two kinds of uncertainty in this network, and represents the most distinctive feature of the network case relative to the value case.

3.4 Directions for Further Analysis

Several substantive questions remain open across both extensions, in roughly increasing order of difficulty.

Existence and uniqueness in the Bayes-Nash setting. As in the baseline case, existence of Bayes-Nash equilibrium follows from the standard route, under the same compactness and continuity assumptions used in Section 2.3. Uniqueness in the network case is more subtle because of the threshold structure: the direct-only corner and the partial-roundabout regime are derived under different active constraints, and how they connect at $\theta = \theta^*$ requires separate analysis. In the value case, uniqueness will likely require a genericity-like condition on the prior F.

Deeper-target case under incomplete information. The shared-edge derivation in Section 3.1 was for a first-target setting (the triangle has both targets as first targets, with the shared edge as an additional route). The full analog of the deeper-target equation, where one target is reached only through another support member, has not been derived in this chapter. It requires an expected version of the indifference between attacking a predecessor and continuing to the deeper target.

Alternative information structures. Both extensions adopt the simplest possible information asymmetries. In the value case, the attacker observes nothing. In the network case, θ is the same for every player. Two natural variants are worth investigating. *Informed attacker:* the attacker observes the full realised type profile while defenders observe only their own type; this is the analog of the Cournot example in Vega-Redondo (2003, Sec. 6.2).

Signal-based observation: the attacker observes a noisy signal correlated with target values (firm size, public-vs-private status), motivated by Chang et al. (2024). The latter is the closest to the empirical cybersecurity context.

Asymmetric priors and stronger value asymmetry. If different targets in different network positions draw from different priors, or if values are far apart in the network-uncertainty case, the equilibrium structure becomes more delicate. Some attack routes drop out of the support, and identifying which routes are used in equilibrium becomes itself part of the computation — analogous to the support-finding iteration in the baseline model. This is a natural extension but introduces additional structure beyond the scope of this introductory treatment.

Larger networks. The triangle is the smallest network where the shared-edge mechanism appears at all. Larger networks bring features the triangle does not have: longer attack paths, deeper-target routing, multi-path coverage, free-highway edges through several non-support nodes. The threshold structure derived in Section 3.2 is specific to the triangle's two-route geometry. In larger networks the threshold might fragment into several thresholds (one per geometric type of route), and the unseen-edge probability would interact with the network's structural features in ways that the triangle cannot reveal. Extending the analysis to networks of more than three nodes is the most important next step.

Combined uncertainty. The two extensions in this chapter are kept separate to isolate the role of each kind of uncertainty. The combined case — values private and network partially observed — has the potential for richer interaction effects but adds substantial complexity to the equilibrium computation. It is the natural setting for cybersecurity applications where neither side observes the other's value at risk nor the full network topology.

Numerical exploration. A first follow-up step is to extend the simulator developed for the baseline model to handle both incomplete-information variants, allowing direct comparison between complete- and incomplete-information equilibria on the small examples used throughout this report. The triangle network analysed above is the natural first test case.

3.5 Conclusion

The edge-based defense model retains tractable structure when either defenders' target values or the network structure itself becomes uncertain. The two extensions developed in this chapter address each kind of uncertainty separately and reveal a clean contrast in how each affects the equilibrium.

Under value uncertainty, each defender's best response depends on the expected behaviour of its neighbors rather than on their actual behaviour, with the expectation taken over a common prior. The resulting Bayes-Nash equilibrium has closed-form expressions in symmetric settings, and the equilibrium quantities depend on moments of the prior rather than on individual realisations. The most striking finding is that uncertainty disrupts the perfect-coordination equilibrium of the complete-information case, inducing both defenders to contribute to shared-edge defense and slightly reducing the attacker's expected payoff.

Under network uncertainty, the analysis on the triangle reveals a distinctive threshold structure. For sufficiently uncertain priors, the attacker disregards roundabout routes entirely, and the defenders respond by leaving the shared edge undefended. Above a critical confidence threshold, the equilibrium gradually returns to the complete-information case. The flat region below the threshold is the most distinctive feature of the network case, and contrasts with the smooth moment-based dependence found in the value case.

These results are derived on the simplest network where the relevant mechanisms appear at all. The triangle's geometry produces clean closed-form results that make the contrast between the two kinds of uncertainty visible, but it also leaves a great deal of structure untested. Extending the analysis to richer networks — and combining the two kinds of uncertainty — is the most natural direction for further work.

Chapter 4

Implementation and Complementary Contributions

Introduction to the Chapter

Chapter 2 established the theoretical results of the modified model under complete information, and Chapter 3 extended them to two settings of incomplete information. This chapter is about everything else: the working tool that was built to verify those results numerically, the skills that the project developed along the way, and the experience of doing research independently for the first time. The chapter is shorter than the analytical chapters, and intentionally so. Theory is the centerpiece of the work; implementation and reflection are what made the theory possible to produce and check.

4.1 From Theory to Working Code

The motivation for building a simulator came early in the project, and it was practical rather than ambitious. Working through the equilibrium of a four-target network by hand takes about an hour. Working through a seven-target network with multiple paths takes the better part of a day, and the answer is then almost impossible to verify without redoing the entire computation from scratch. By the time the model needed to be tested on networks large enough to demonstrate the multi-path effect clearly, hand calculation had become a bottleneck.

A simulator removes that bottleneck. It also does something else, which turned out to be more important. It makes the theory testable in a way that printed equations are not. Once the model is in code, every claim made in Chapter 2 can be checked against numerical output. The condition that target values strictly increase along an attack path, the equilibrium equations with the multi-path factor m_i, the cost-savings result on lines, the disappearance of Braess's paradox — each of these was verified by running the simulator on representative networks and confirming that the numerical output matched the analytical prediction. When something failed to match, the disagreement always pointed to a real error somewhere, in the code or in the math or in my own understanding. This kind of feedback loop is hard to produce any other way.

The choice of R Shiny as the platform was deliberate. R has strong libraries for graph manipulation (`igraph`) and visualization (`ggraph`), which made the network display layer straightforward to build. Shiny adds a web interface on top, so the simulator can be deployed publicly and accessed from any browser without installation. The complete tool is hosted at https://chukwudi-ogbuta.shinyapps.io/blochnetwork/ and runs without any setup on the user's side.

It is worth noting that the computational side of Bloch's original model was independently addressed by Kaźmierowski and Dziubiński (2023), who established that a Nash equilibrium of the node-based game can be computed in polynomial time — $O(n^4)$ with respect to the number of nodes — through an algorithm based on network reduction and the identification of a class of nodes called *linkers* that are never attacked in any equilibrium. Their work resolves an open question left by Bloch et al. regarding the computational tractability of finding the equilibrium support. The simulator described in this chapter is a different kind of contribution. It is not a formal complexity result but an interactive tool, and it covers two things their algorithm does not: the edge-based modification developed in Chapter 2, and the cooperative version of both models. The two contributions are complementary — a formal complexity treatment on the one hand, an interactive verification and experimentation environment on the other.

4.2 How the Simulator Works

The application is organized into three tabs that follow the natural flow of investigation: build a network, solve its equilibrium, and run experiments on it.

The Network Builder

The first tab lets the user construct a network from scratch. The user chooses the network type (line, star, or general), specifies the number of targets, and enters a value for each target. For general networks, an edge selector lets the user click on which pairs of nodes are connected. The result is rendered as a graph, with nodes colored by value and the attacker shown in red.

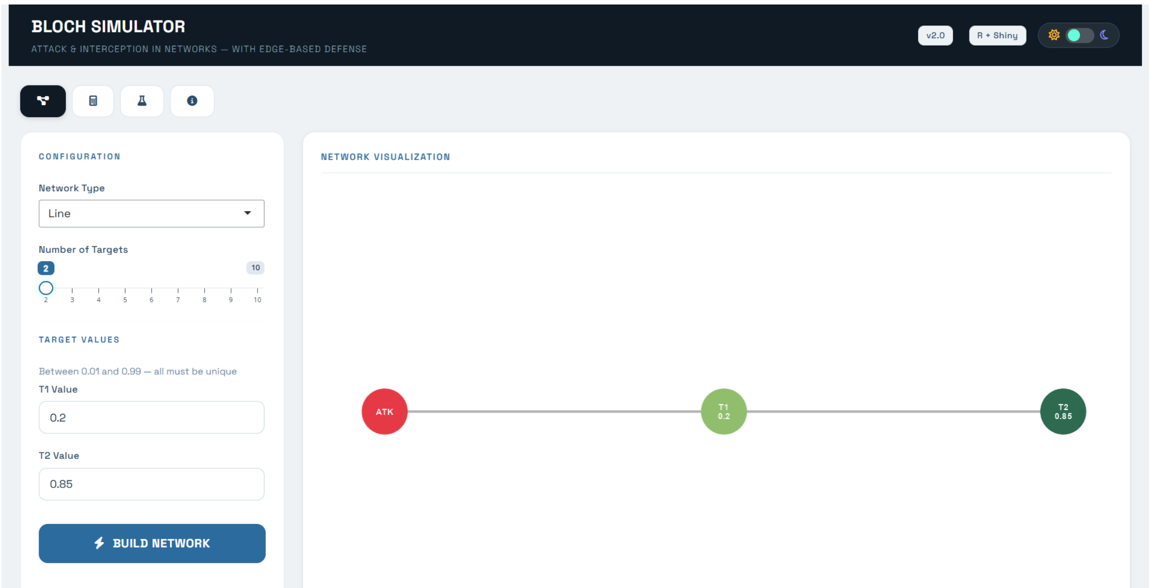


Figure 4.1: The Network Builder tab. The user specifies the network structure and target values; the simulator displays the resulting graph.

The builder enforces basic consistency checks before allowing the network to be used: target values must be unique (the genericity assumption), every target must be reachable from the attacker, and no isolated nodes are allowed. These checks catch mistakes before they propagate into the solver.

The Equilibrium Solver

The second tab is where the analytical results meet the keyboard. The user selects a defense model (node-based as in Bloch's original formulation, or edge-based as in the modification), chooses between decentralized and cooperative defense, and runs the solver. The simulator returns the attacker's expected utility U, the equilibrium support, the attack probabilities q_i, and the per-edge investments. A side-by-side comparison view runs both defense models on the same network and presents the results next to each other.

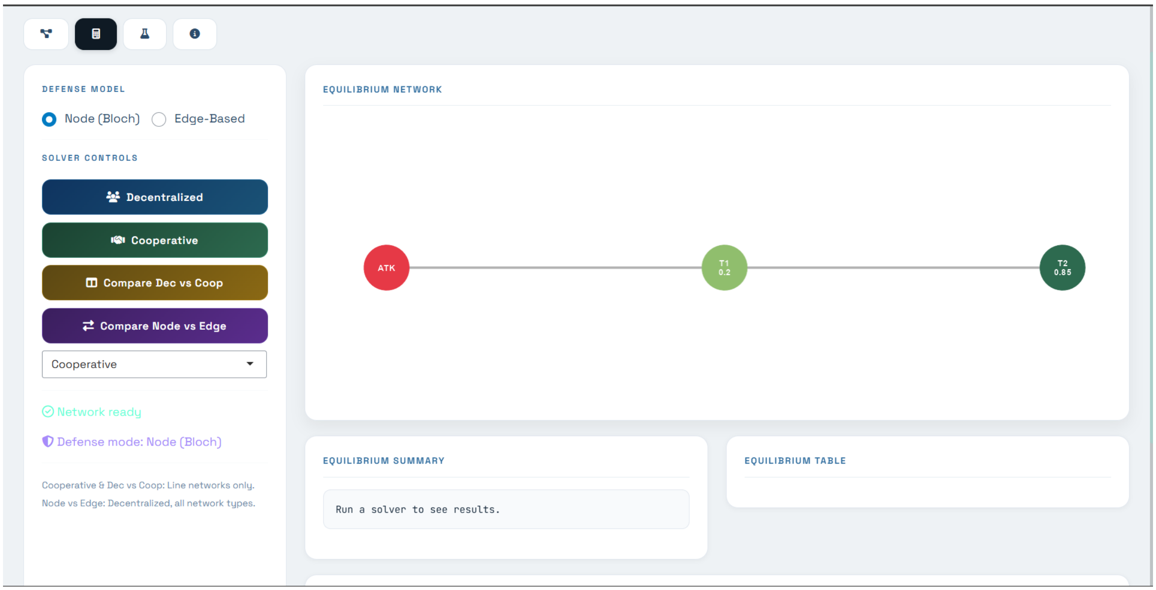


Figure 4.2: The Equilibrium Solver tab. The user picks a defense model and solver type; the simulator returns the equilibrium values, attack probabilities, and per-edge investments.

The solver implements the iterative support-reduction procedure detailed in the Appendix (summary table in Chapter 2). It starts by assuming all targets are in the support, computes the equilibrium under this assumption, and removes any target whose attack probability comes out non-positive. It repeats until every remaining target has a strictly positive attack probability. For the edge-based model, the solver computes the multi-path factor m_i for each surviving target by counting the free-highway incoming edges, and applies the modified equilibrium equations. A full step-by-step walkthrough of this iteration is given in the Appendix, where each example's support is identified explicitly round by round.

The Experimentation Tab

The third tab implements the comparative statics from Chapter 2 directly. The user picks an experiment from a menu — changing a target's value, adding a new connection, or removing a target from the attacker's consideration — specifies the parameters, and runs it. The simulator solves the equilibrium twice, once on the original network and once on the modified version, and presents the results side by side along with an impact summary that quantifies the change in attacker utility.

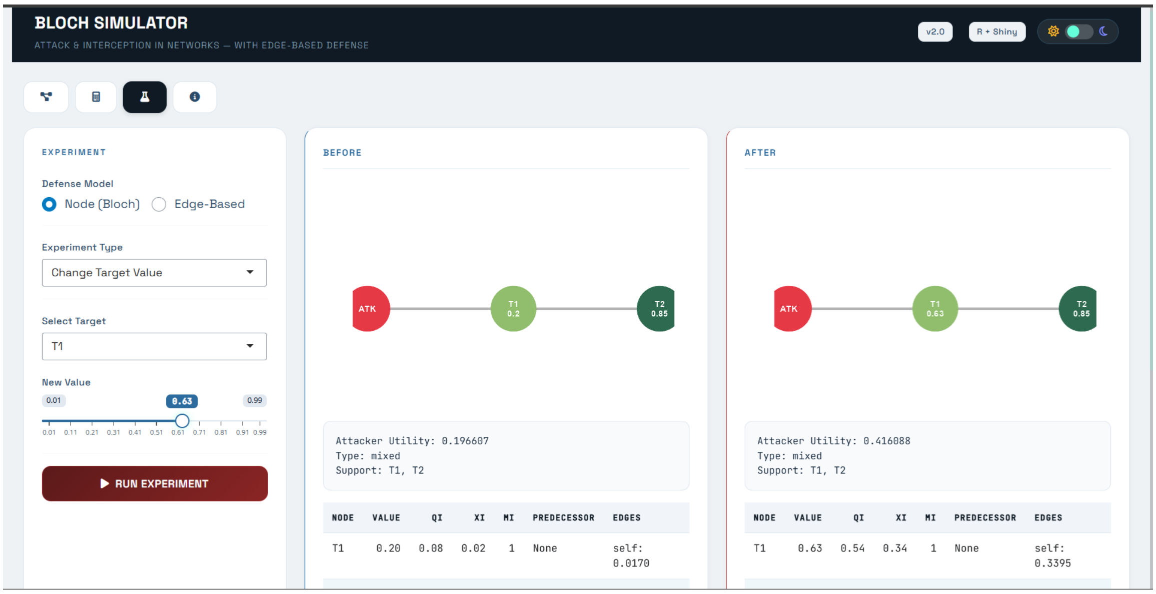


Figure 4.3: The Experimentation tab. The user perturbs the network and sees how the equilibrium shifts.

This tab is the part of the simulator most directly connected to the report's analytical chapter. It turns the comparative statics into something the user can run rather than just read about.

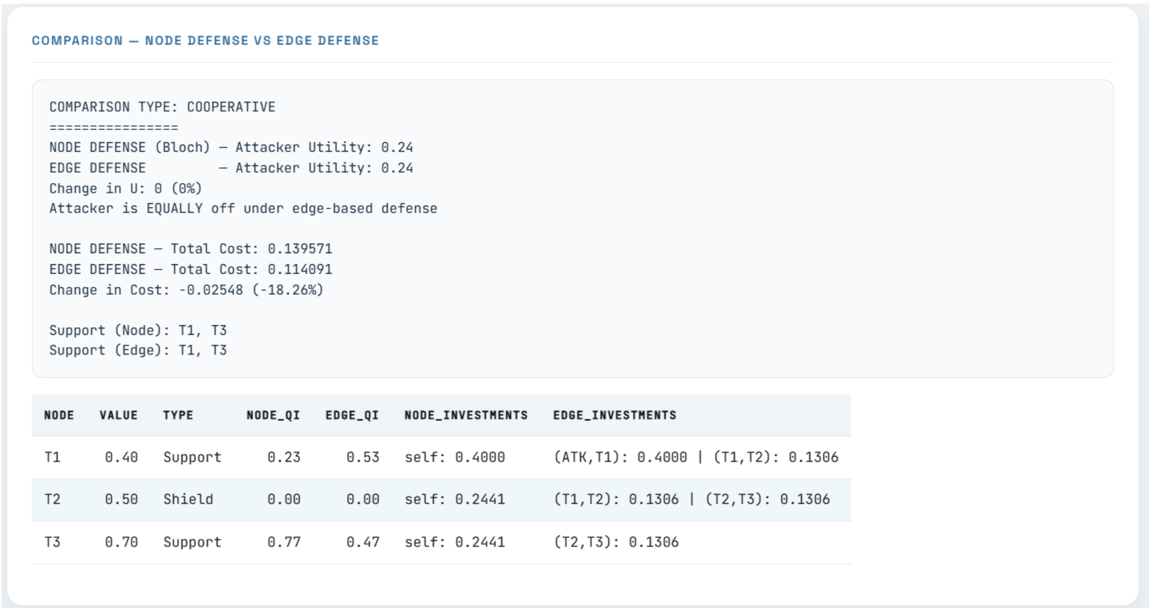
4.3 Numerical Verification of the Examples in Chapter 2

The two worked examples in Chapter 2 served as the primary test cases for the simulator. Their numerical predictions can be confirmed directly in the application.

Example 1: Cooperative Comparison on a Line

The line network with values $(0.40, 0.50, 0.70)$ was constructed in the Network Builder and solved using the cooperative comparison feature. The output reproduced the values stated in Chapter 2: attacker utility $U = 0.24$ in both defense models, with total cost falling from 0.1396 under node defense to 0.1142 under edge defense, a saving of 18.2%. The equilibrium table displays the per-edge investments under edge mode, making explicit that node T_2 invests on each of its two adjacent edges and that the total cost on each interior edge reflects two endpoints contributing equally.

Figure 4.4: Example 1: a line network of three targets with values $(0.40, 0.50, 0.70)$.



COMPARISON — NODE DEFENSE VS EDGE DEFENSE

```
COMPARISON TYPE: COOPERATIVE
================
NODE DEFENSE (Bloch) — Attacker Utility: 0.24
EDGE DEFENSE         — Attacker Utility: 0.24
Change in U: 0 (0%)
Attacker is EQUALLY off under edge-based defense

NODE DEFENSE — Total Cost: 0.139571
EDGE DEFENSE — Total Cost: 0.114091
Change in Cost: -0.02548 (-18.26%)

Support (Node): T1, T3
Support (Edge): T1, T3
```

NODE	VALUE	TYPE	NODE_QI	EDGE_QI	NODE_INVESTMENTS	EDGE_INVESTMENTS
T1	0.40	Support	0.23	0.53	self: 0.4000	(ATK,T1): 0.4000 | (T1,T2): 0.1306
T2	0.50	Shield	0.00	0.00	self: 0.2441	(T1,T2): 0.1306 | (T2,T3): 0.1306
T3	0.70	Support	0.77	0.47	self: 0.2441	(T2,T3): 0.1306

Figure 4.5: Example 1 solution: side-by-side cooperative comparison under node and edge defense.

The match between the analytical prediction and the simulator output confirms the cost-savings result derived in Chapter 2. Same level of effective interception at every position, lower total cost under edge defense, and the difference traceable directly to the convexity of the quadratic cost function.

Example 2: Decentralized Comparison on a General Network

The four-target general network with values $(0.20, 0.25, 0.50, 0.70)$ tests the multi-path weakness result. Target T_4 has two incoming edges, one through T_1 and one through T_2, both free-highway since T_1 and T_2 are not in the support. The simulator confirms the predicted equilibrium: under node defense, $U = 0.4020$; under edge defense, $U = 0.4773$. The change in U of $+18.7\%$ reflects the structural weakness introduced by the multi-path factor $m_4 = 2$. Attack probability on T_4 rises from 61% to 91%, and the investment at T_3 collapses from 0.196 to 0.045 as T_3 becomes nearly indifferent to attack at the higher equilibrium U.

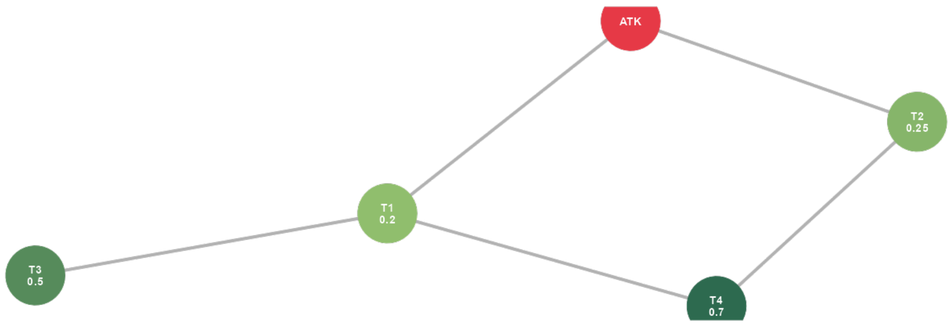
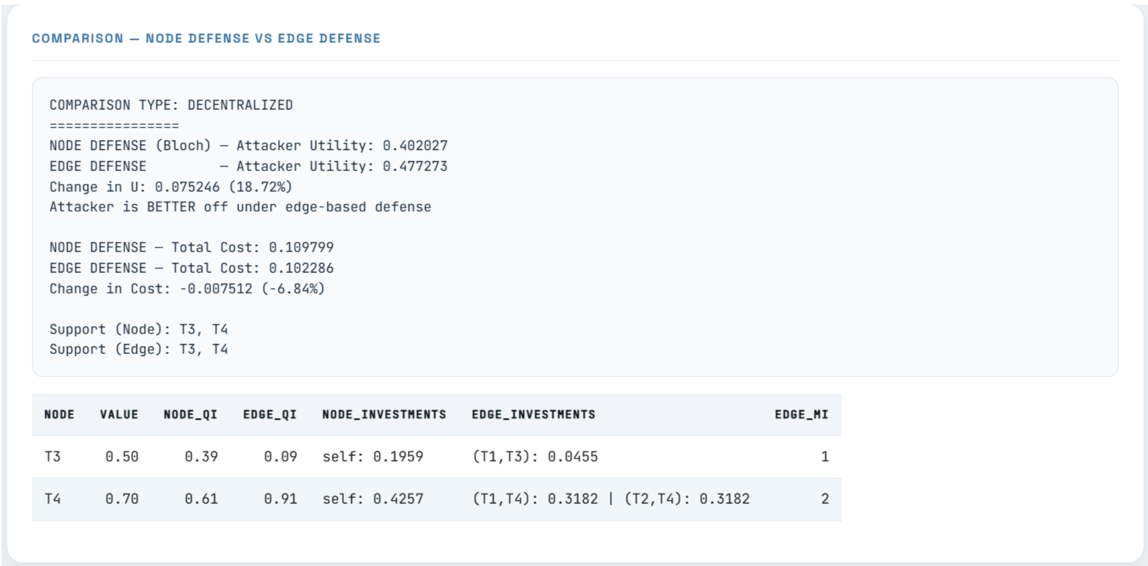


Figure 4.6: Example 2: a general network with T_4 reachable through two distinct paths.



COMPARISON — NODE DEFENSE VS EDGE DEFENSE

```
COMPARISON TYPE: DECENTRALIZED
================
NODE DEFENSE (Bloch) — Attacker Utility: 0.402027
EDGE DEFENSE         — Attacker Utility: 0.477273
Change in U: 0.075246 (18.72%)
Attacker is BETTER off under edge-based defense

NODE DEFENSE — Total Cost: 0.109799
EDGE DEFENSE — Total Cost: 0.102286
Change in Cost: -0.007512 (-6.84%)

Support (Node): T3, T4
Support (Edge): T3, T4
```

NODE	VALUE	NODE_QI	EDGE_QI	NODE_INVESTMENTS	EDGE_INVESTMENTS	EDGE_MI
T3	0.50	0.39	0.09	self: 0.1959	(T1,T3): 0.0455	1
T4	0.70	0.61	0.91	self: 0.4257	(T1,T4): 0.3182 | (T2,T4): 0.3182	2

Figure 4.7: Example 2 solution: side-by-side decentralized comparison under node and edge defense.

The simulator output reproduces every quantity reported in Chapter 2's Table 2.2, providing an independent numerical check on the analytical derivation.

4.4 A Worked Comparative Experiment

To illustrate how the experimentation tab connects the theoretical analysis to interactive use, consider the following comparative-statics experiment on the general network from Example 2. The original network has T_4 reachable through two distinct free-highway edges, giving multi-path factor $m_4 = 2$. What happens if a new connection is added between T_3 and T_4?

Adding this edge changes the structure of attacks on T_4. Previously T_4 had two incoming free-highway edges; now it has three, since T_3 is also not in the support of the modified equilibrium and the path ATK $\to T_1 \to T_3 \to T_4$ becomes a third free-highway route. The multi-path factor m_4 rises from 2 to 3.

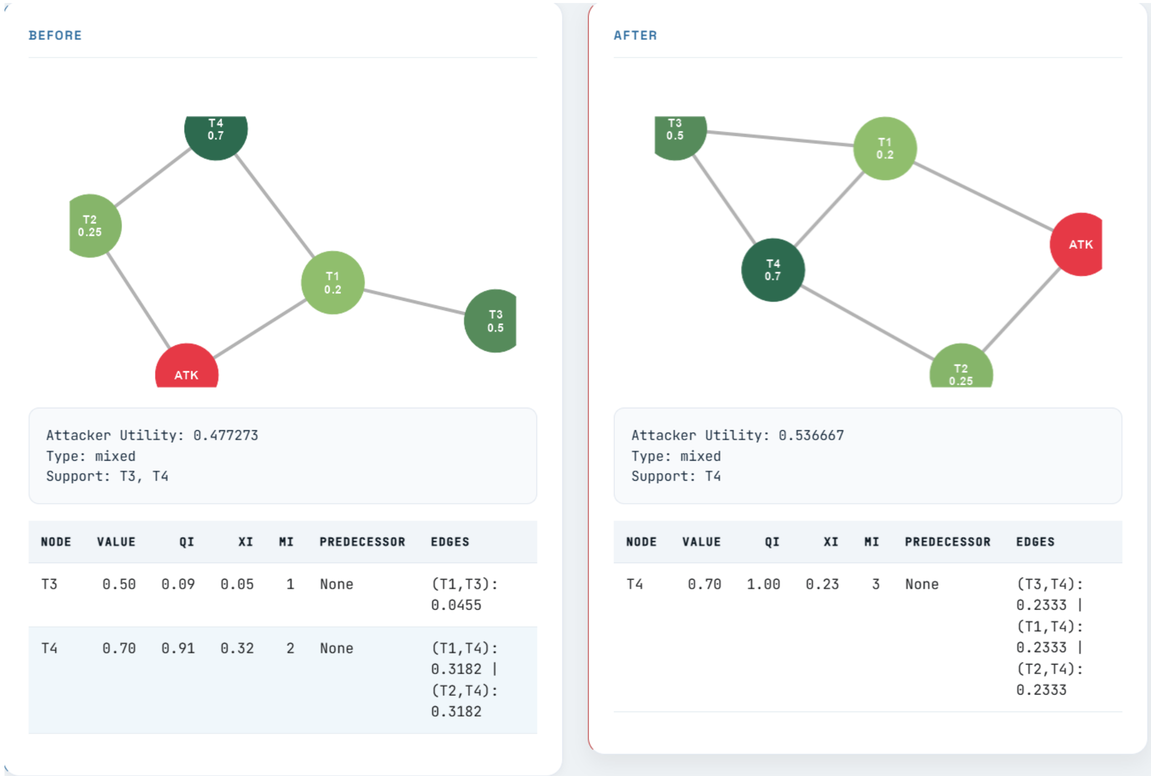


Figure 4.8: The comparative experiment: adding edge (T_3, T_4) to the general network. The simulator solves the equilibrium before and after the modification.

The simulator output makes the consequences explicit. The attacker's expected utility rises from 0.4773 to 0.5367, an increase of about 12.4%. Target T_3, which had a small but positive attack probability before the modification, drops out of the support entirely after the new edge is added. All attack probability concentrates on T_4, which now defends three incoming edges at a per-edge investment of 0.2333 each. The total cost picture shifts as well: T_3 no longer pays anything, while T_4's burden grows because it must defend an additional edge.

The reasoning behind the result is the same logical chain identified in Chapter 2. More incoming free-highway edges to a target means each edge is structurally weaker, since the per-edge investment formula $x = 1 - U/b_i$ is unchanged but must now be maintained on more edges. The aggregate effect raises U, which pushes lower-value targets out of the support, which in turn concentrates the attacker's probability on the multi-path target. The experiment thus confirms the theoretical claim that adding a connection to a multi-path target unambiguously benefits the attacker — a result that distinguishes the edge-based model from Bloch's, where Braess-type paradoxes are possible.

This experiment is the kind of investigation that would have taken hours to do by hand and now takes about ten seconds in the simulator. The interactive format also makes it easier to explain the result to others. Showing a supervisor the before-and-after view conveys the dynamic in a way that a static table of numbers does not.

4.5 Skills and Working Methods

Beyond the theoretical and computational outputs, the internship developed a set of skills that I want to record honestly rather than as a list of bullet points.

Technical Skills

The most concrete additions to my technical toolkit were the formal mathematics of game theory and the corresponding software workflow. Before this internship I had a graduate-level exposure to game theory through coursework, but I had never derived a Nash equilibrium from first principles, never written out a proof of existence using the Debreu-Fan-Glicksberg framework, and never worked through comparative statics by implicit differentiation on an equilibrium equation. Doing each of these for the first time, with corrections from a supervisor who has spent his career in game theory, taught me how the formal apparatus actually works.

The software side developed similarly. I had used R for data analysis, but not for building applications. Shiny was new, the deployment workflow was new, and the integration of `igraph` for network logic with `ggraph` for visualization required a pattern of separating computation from display that I had not encountered before. Writing the simulator also forced me to internalize the structure of the equilibrium algorithm. There is a difference between understanding a procedure on paper and being able to write code that executes it correctly; the simulator was the place where that difference became visible to me.

LaTeX was familiar from earlier coursework, but this project pushed it further. The proofs document, the presentation slides, and the report itself all required structures more elaborate than anything I had written before — proper theorem environments, bibliographies, cross-references, tables with consistent formatting. The discipline of producing professional academic documents in LaTeX is now a habit rather than an effort.

Analytical Skills

The most important analytical lesson was about how research questions are actually structured. Going in, I imagined the work would proceed in a straight line: take the model, change one feature, derive the consequences. In practice the work involved a constant tension between making changes that were small enough to remain tractable and making changes that were substantive enough to be interesting. Several variations were considered and abandoned because their consequences became impossible to characterize cleanly. The variation that became the focus of the report was selected as much for its analytical clarity as for its conceptual interest.

Working through this taught me to distinguish between a question that is hard because it is genuinely deep and a question that is hard because it is poorly posed. A good model adjustment is one where each change has a traceable consequence, and where the resulting analysis can be carried out without resorting to numerical methods or special-case arguments. The edge-based modification fits this description; several alternatives I tried did not.

Working Methods and Cross-Validation

A practical habit that the internship reinforced was the use of multiple tools to cross-check results. Throughout the project I worked with Claude (for writing, derivation, and code), DeepSeek (for paper analysis and second opinions on derivations), and the simulator I built (for numerical verification). When all three agreed on a result, I had high confidence in it. When they disagreed, the disagreement itself was informative — one of them was wrong, and the discipline of figuring out which one taught me more than getting the right answer the first time would have. There were several occasions during the project where one of the tools produced an answer that turned out to be incorrect; the cross-validation caught these before they made it into the report.

This habit generalizes. Trusting any single source — including one's own first calculation — is fragile. Building the practice of independent verification, whether through a second tool, a different method of derivation, or a numerical check, is one of the most useful working methods I take from this internship.

4.6 A Reflection on the Process

The largest difference between this internship and the work I had done before was the absence of a fixed structure. A coursework assignment comes with a question, an expected method, and a known approximate length of time to completion. A research internship is different. The question itself was open-ended at the start, and a substantial part of the early work was figuring out what question to actually answer. Several of the model variations I considered in the first weeks turned out to be either too modest to be interesting or too ambitious to be tractable. Settling on the edge-based modification as the focus took several weeks of conversation with my supervisor and several false starts.

Working independently for extended periods also developed a discipline I had not built before. Without a deadline at the end of every week, the temptation to delay difficult parts is strong. The countermeasure I found most useful was setting myself small, specific milestones — a single proof to write, a single example to compute, a single section of code to refactor — and treating each of them as a unit of work that could be completed in a sitting. The cumulative effect of this approach was that the report grew gradually rather than in panicked bursts.

A related lesson was about how to use a supervisor effectively. Early in the internship I tended to bring my supervisor finished work for review. This was inefficient: by the time the work was finished, the underlying choices had already been made, and any feedback came too late to change them. Later in the internship I learned to bring questions and partial work earlier, when the direction was still mutable. The conversations about the model setup, the edge-based modification, the role of m_i, and the joint-cost variation were all most productive when I came in with a half-formed idea rather than a polished one. This is a habit I expect to carry forward.

Conclusion to the Chapter

The simulator described in this chapter is not a side project to the theoretical work but a continuous part of it. Every result in Chapter 2 was checked numerically, and several errors in early drafts of the analysis were caught by the simulator before they made it into writing. The skills developed during the internship — in formal mathematics, in software, in academic writing, in working independently — are the practical companion to the theoretical contribution. Together they constitute what the internship produced.

General Conclusion

Summary of the Work

This report set out to answer a single question: when defense in a network attack game is moved from the locations to the connections between them, how does the strategic situation change? The starting point was the model of Bloch, Chatterjee and Dutta (2023), whose computational side has since been resolved by Kaźmierowski and Dziubiński (2023), and the inspiration for the modification came from Mavronicolas et al. (2008), who showed in a discrete setting that placing defense on edges produces equilibria with a recognizably different structure.

The work proceeded in four connected parts. The first part, presented in Chapter 1, situated the contribution within two parallel traditions in the literature on network security games: the node-defense tradition that Bloch's model belongs to, and the smaller edge-defense tradition that has so far worked only with discrete, centralized formulations. The gap between them — a continuous, decentralized model with defense on edges — is what this report attempts to fill.

The second part, presented in Chapter 2, built the modified model from the ground up and derived its formal results. The model retains Bloch's continuous-investment framework but places each defender's investment on the edges incident to its node rather than at the node itself. Each edge is thus jointly defended by its two endpoints, with survival through the edge given by the product of the two endpoints' interception probabilities. From this single change, several consequences follow. Some elements of Bloch's analysis carry over: the equilibrium exists, is unique under the standard genericity assumption, and the equations governing deeper targets remain the same. Other elements change in ways that are structurally informative. Targets reachable through multiple incoming free-highway edges are penalized by a factor m_i that captures the number of such edges; this multi-path weakness raises the attacker's expected utility and concentrates attack probability on multi-path targets. Edges shared by two support members exhibit strategic substitutes, with each defender's marginal incentive to invest declining in the neighbor's investment. The cooperative version of the model produces two distinctive results: investments are symmetric within each edge, and on line networks the same level of attacker harm is achieved at strictly lower defense cost than under Bloch's node-based cooperative.

The third part, presented in Chapter 3, extended the analysis to two distinct kinds of incomplete information. The first extension treats target values as private types drawn from a common prior, with defenders observing only their own value and the attacker operating under the prior alone; the resulting Bayes-Nash equilibrium admits a closed-form characterisation in symmetric settings. The second extension fixes target values as common knowledge but introduces partial observation of the network structure, with each player observing only its own incident edges and forming beliefs about the rest; the equilibrium on the triangle has a distinctive threshold structure in which network uncertainty becomes absorbing below a critical confidence level. Together the two extensions isolate the role of each kind of uncertainty and produce a clean contrast: value uncertainty hurts the attacker through coverage spreading on shared edges, while network uncertainty hurts the attacker through route abandonment. Both extensions are developed on the triangle network as a first picture; their generalisation to larger networks is identified as the main next step.

The fourth part, presented in Chapter 4, made the theory testable. An interactive R Shiny simulator was built to verify the analytical results numerically, support comparative experimentation, and produce the comparison tables that appear in Chapter 2. The simulator confirms each numerical claim made in the analytical chapter and allowed several errors in early drafts of the analysis to be caught before they made it into the final report.

Limitations

Several limitations of the work deserve acknowledgment. The cooperative analysis is fully developed for line networks and extends cleanly to two-branch trees, in which two lines fan outward from the attacker; a worked example is given in the appendix (Section A.4). In both topologies the analysis remains tractable because each target is reached through a unique path from the attacker, so the coordinator's first-order conditions decompose: branches couple only through the common equilibrium payoff U. The structure breaks for networks in which a single target is reached through multiple paths in the network interior, such as the general network of Example 2 in Chapter 2. There, an investment on an upstream shield edge influences attacks heading to two different downstream targets simultaneously, so the first-order conditions no longer separate into independent branch subproblems. The full system can still be set up and solved numerically, but a clean closed-form characterisation analogous to the line and two-branch tree is not available. The strategic substitutes mechanism on shared edges raises questions about uniqueness and stability of the interior equilibrium that this report does not resolve — the existence and characterization of equilibria when both endpoints of an edge belong to the support is established, but a full game-theoretic analysis of how the two endpoints arrive at a stable interior solution remains open. The joint-cost variation discussed at the end of Chapter 2 was sketched rather than fully analyzed; its detailed treatment is a natural extension that would require its own dedicated study.

The report also restricts attention to a single attacker. Multi-attacker extensions, in which several adversaries operate simultaneously on the same network, would substantially complicate both the equilibrium structure and the cooperative analysis, and were beyond the scope of this work.

Directions for Future Research

Several directions for future work follow naturally from the analysis presented here. A characterisation of cooperative defense on deeper trees — structures in which each target is still reached through a unique path from the attacker, but the branching extends beyond the simple two-branch case treated here — is the natural next step and would likely admit the same kind of branch-decomposition argument. The harder question is cooperative defense on networks with multi-path targets, where the first-order conditions no longer decompose into independent subproblems and a clean analytical treatment is not yet available; identifying the structural assumptions under which such networks can be characterised, or developing alternative methods of analysis, is a substantive research programme in itself. The asymmetric joint-cost variation, in which the two endpoints of a shared edge contribute under a single combined cost function rather than separable individual costs, presents a clean variation that the framework can handle and would amplify the strategic substitutes effect. Chapter 3 developed a first treatment of incomplete information in two specialised settings; extending those results to larger networks, combining the two kinds of uncertainty, and treating richer information structures (informed attackers, signal-based observation) are natural follow-ups that build directly on the framework developed there. Multi-attacker extensions, network design questions (where does the defender place new connections to minimize attacker payoff?), and stability analysis of the strategic-substitutes interior equilibrium each represent further substantive research programs that build on the modification developed here.

A practical direction, complementary to the theoretical extensions, is the application of the model to specific resource-management systems. The motivating examples in this report — illegal logging in transboundary forests, illegal fishing in shared waters, and illegal waste trafficking across border corridors — have data structures rich enough to support empirical calibration of the model's parameters. A full case study on any one of these systems would test whether the qualitative predictions of the model survive contact with real-world data and whether the cost-savings result on lines extends meaningfully to more complex network shapes.

Personal Reflection

For me as a student, the most valuable outcome of this internship was not any single technical result. It was the experience of working on a research problem with no fixed structure, no predetermined method, and no guaranteed answer. Coursework had prepared me for exercises with known solutions; this internship taught me what it feels like to work on something where the question itself was open at the start, where several promising directions had to be abandoned along the way, and where the right answer had to be earned rather than given.

The discipline of cross-validating results across multiple tools, the habit of bringing partial work to a supervisor early rather than finished work too late, the practice of building a numerical check for every analytical claim — these are working methods I expect to carry forward into whatever I do next. The internship also clarified for me that the field of economic intelligence, which sits at the intersection of strategic analysis, formal modeling, and quantitative tools, is a direction I want to continue pursuing.

I am grateful to Professor Sylvain Béal for the opportunity to undertake this work and for the supervision that shaped it, and to CRESE and the BDEEM program for the institutional and intellectual environment that made it possible.

Appendix A

Detailed Calculations

The formal proofs of the results stated in Chapter 2 are given in the body of that chapter, immediately after each statement. This appendix collects the longer step-by-step calculations that would interrupt the flow of the main text: the full working behind the two numerical examples, the working behind the comparative experiment reported in Chapter 4, and the algebra confirming the cost-savings inequality. The notation is that of Chapter 2: b_i is the value of target i, U the equilibrium attacker payoff, q_i the attack probability on target i, $x_{i,e}$ the investment by node i on edge e, and m_i the multi-path factor of a first target.

A.1 Worked Calculation for Example 1: The Line Network

The network is ATK–T_1–T_2–T_3 with values $b_1 = 0.40$, $b_2 = 0.50$, $b_3 = 0.70$, solved under cooperative defense in both the node-based and edge-based formulations.

Identifying the support: the iteration in full

We first ask which targets the attacker actually strikes under cooperative defense. The procedure is the same elimination used for Example 2: start with all targets as candidates, solve, and drop any candidate whose attack probability is non-positive.

Round 1: all three targets as candidates. On the line, the targets are met in the order T_1, T_2, T_3 with increasing values $0.40 < 0.50 < 0.70$. As a diagnostic for whether the lowest candidate T_1 is actually the first attacked target, we use the line test of Bloch's Proposition 2: T_1 is the first attacked target only if

$$\sum_{j=2}^{3} \left(1 - \frac{b_{j-1}}{b_j}\right) \frac{b_{j-1}}{b_1\, b_j} \;\leq\; 1.$$

Computing the two terms:

$$\begin{aligned} j=2: &\quad \left(1-\frac{0.40}{0.50}\right)\frac{0.40}{0.40\times 0.50} = (0.2000)(2.0000) = 0.4000, \\ j=3: &\quad \left(1-\frac{0.50}{0.70}\right)\frac{0.50}{0.40\times 0.70} = (0.2857)(1.7857) = 0.5102. \end{aligned}$$

The sum is $0.4000+0.5102 = 0.9102 \leq 1$, so T_1 is the first attacked target, and all targets from T_1 onward are candidates: the decentralized support is $\{T_1, T_2, T_3\}$.

From decentralized to cooperative: T_2 becomes a shield. The decentralized support is the starting point for the cooperative problem, but the cooperative coordinator has an option no individual defender has: it can deliberately drop a target from the attack support and protect it purely as a *shield* for the targets behind it. Dropping T_2 means the coordinator still invests on the positions around T_2 — because T_2 lies on the only route to T_3 — but arranges investments so that the attacker is left indifferent only between T_1 and T_3 and has no incentive to strike T_2. Solving the cooperative problem both ways (with T_2 in the attack support, and with T_2 as a shield) and comparing total cost, the shield configuration is the cooperative optimum. The cooperative support is therefore $\Delta = \{T_1, T_3\}$, with T_2 defended but not attacked.

This shield phenomenon is exactly the "gap in the support" that Bloch et al. (2023) identify in their cooperative analysis: under coordination, the attacked set need not be a contiguous run of targets along the line.

Node-based cooperative equilibrium

With support $\{T_1, T_3\}$ and T_2 as a shield, the coordinator chooses node investments x_1, x_2, x_3 to minimize total expected loss. We work through the first-order conditions explicitly.

First target T_1. The attacker reaches T_1 across the entrance edge alone, so the attacker's payoff from T_1 is $b_1(1-x_1)$, and in equilibrium this equals the common value U. The coordinator's first-order condition for x_1 balances the marginal reduction in expected loss against the marginal cost x_1; carrying it out gives

$$x_1 = b_1 = 0.40,$$

and therefore

$$U = b_1(1-x_1) = 0.40 \times 0.60 = 0.24.$$

Shield T_2 and target T_3. The attacker reaches T_3 only by passing T_2, so the survival to T_3 is $(1-x_1)(1-x_2)(1-x_3)$ and the attacker's payoff from T_3 is $b_3(1-x_1)(1-x_2)(1-x_3)$. The coordinator's first-order conditions for x_2 and x_3 enter the loss function symmetrically — each appears once as a factor $(1-x)$ in the survival product and once as a quadratic cost — so they yield $x_2 = x_3$. Writing the common value as z and using the simulator's solution of the cooperative system,

$$x_2 = x_3 = 0.2441.$$

Attack probabilities. With U and the investments fixed, the attacker's mixing probabilities r_1 and r_3 are pinned down by the coordinator's first-order conditions and the requirement $r_1 + r_3 = 1$. The cooperative system gives

$$r_1 = 0.2313, \qquad r_3 = 0.7687.$$

Total cost. Summing the quadratic cost of each node investment,

$$C^{\text{node}} = \frac{x_1^2 + x_2^2 + x_3^2}{2} = \frac{0.40^2 + 0.2441^2 + 0.2441^2}{2} = \frac{0.1600 + 0.0596 + 0.0596}{2} = 0.1396.$$

Edge-based cooperative equilibrium

Under edge defense the same three positions must be covered, but every interior edge is now paid for from both endpoints. By the symmetry-within-edges result, the two endpoints of each interior edge invest equally. To reproduce the same effective interception as the node model at each interior position, the per-endpoint investment z on an interior edge must satisfy $(1-z)^2 = 1 - x^{\text{node}}$, where x^{node} is the corresponding node investment. For positions 2 and 3, $x^{\text{node}} = 0.2441$, so

$$(1-z)^2 = 1 - 0.2441 = 0.7559 \quad \Longrightarrow \quad z = 1 - \sqrt{0.7559} = 0.1307.$$

The entrance edge (ATK, T_1) has only one investor, so its investment is unchanged at 0.4000. The equilibrium value is the same, $U = 0.2400$, because the effective interception at every position is unchanged; the cooperative attack probabilities are $r_1 = 0.5264$ and $r_3 = 0.4736$.

The total cost sums the entrance edge plus the two interior edges, each interior edge carrying two equal investments of 0.1307:

$$C^{\text{edge}} = \frac{0.4000^2}{2} + 2 \cdot \frac{0.1307^2}{2} + 2 \cdot \frac{0.1307^2}{2} = 0.0800 + 0.0171 + 0.0171 = 0.1142.$$

The comparison

Both formulations deliver $U = 0.2400$. The node-based total cost is 0.1396; the edge-based total cost is 0.1142. The saving is

$$\frac{0.1396 - 0.1142}{0.1396} = 0.182,$$

that is, 18.2%. The saving comes entirely from the two interior positions, where edge defense replaces one investment of 0.2441 (cost 0.0298) with two investments of 0.1307 (combined cost 0.0171).

A.2 Worked Calculation for Example 2: The General Network

The network has the attacker connected to T_1 and T_2; T_1 connected to T_3 and T_4; and T_2 connected to T_4. Values are $b_1 = 0.20$, $b_2 = 0.25$, $b_3 = 0.50$, $b_4 = 0.70$. The comparison is under decentralized defense.

Identifying the support: the iteration in full

The support is found by an elimination procedure. We begin with every target as a candidate, solve the equilibrium constraint $\sum_i q_i(U) = 1$, and check that every candidate receives a strictly positive attack probability. Any candidate whose attack probability comes out non-positive cannot be in the support and is removed; the procedure is then repeated on the smaller candidate set, and continues until every remaining candidate has a strictly positive attack probability. We show every round explicitly for the node-based model.

Round 1: all four targets as candidates. The targets T_1 and T_2 are directly reachable from the attacker, so as candidates they are first targets; T_3 is reached through T_1, and T_4 through T_1 and T_2. A target's classification as first or deeper is determined by the attacker's optimal route, not by the topology of the network — T_2 for instance has a roundabout route ATK–T_1–T_4–T_2 that would make it a deeper target, but this route faces strictly more defense than the direct edge (ATK, T_2) and is never used in equilibrium. If all four candidates were in the support, the first-target equation $q_i = \frac{1}{b_i}(1 - U/b_i)$ would apply to T_1 and T_2, and the deeper-target equation to T_3 and T_4. But the first-target equation already exposes the problem. For T_1,

$$q_1 = \frac{1}{b_1}\left(1 - \frac{U}{b_1}\right) = \frac{1}{0.20}\left(1 - \frac{U}{0.20}\right),$$

which is positive only if $U < b_1 = 0.20$. The equilibrium value U is the payoff the attacker can secure at every supported target; it cannot be as low as 0.20 when targets worth 0.50 and 0.70 are available, since attacking those (even defended) yields more. So q_1 is forced non-positive: T_1 cannot be in the support. The identical argument applies to T_2, whose attack probability is positive only if $U < b_2 = 0.25$. Both T_1 and T_2 are removed.

Round 2: candidates $\{T_3, T_4\}$. With T_1 and T_2 removed, they invest zero and become transparent: the attacker passes through them at no cost. T_3 is now reached from the attacker along a route (ATK $\to T_1 \to T_3$) whose only intermediate node, T_1, invests zero — so T_3 is a first target. The same holds for T_4 along ATK $\to T_1 \to T_4$ and ATK $\to T_2 \to T_4$. Both candidates are therefore first targets, and the first-target equation applies to each. We solve $q_3 + q_4 = 1$ in the next subsection and find $U = 0.4020$ with $q_3 = 0.3920 > 0$ and $q_4 = 0.6081 > 0$. Both attack probabilities are strictly positive, so the procedure terminates: the support is $\Delta = \{T_3, T_4\}$.

Consistency check on the dropped targets. With the solved value $U = 0.4020$, we confirm that T_1 and T_2 were correctly excluded: $1 - U/b_1 = 1 - 0.4020/0.20 = -1.010 < 0$ and $1 - U/b_2 = 1 - 0.4020/0.25 = -0.608 < 0$. Both would give negative attack probabilities, confirming they do not belong in the support.

The same support $\Delta = \{T_3, T_4\}$ is obtained in the edge-based model by the identical iteration. Because T_1 and T_2 are dropped, the edges they sit on carry no defense, so they become free-highway edges. Target T_4 is then reached by two free-highway edges — one through T_1, one through T_2 — giving $m_4 = 2$. Target T_3 is reached by the single free-highway edge through T_1, giving $m_3 = 1$.

Node-based equilibrium

In the node-based model both T_3 and T_4 are first targets, each with the Bloch first-target equations

$$q_i = \frac{1}{b_i}\left(1 - \frac{U}{b_i}\right), \qquad i \in \{3, 4\}.$$

Imposing $q_3 + q_4 = 1$:

$$\frac{1}{0.50}\left(1 - \frac{U}{0.50}\right) + \frac{1}{0.70}\left(1 - \frac{U}{0.70}\right) = 1.$$

Expanding,

$$2(1 - 2U) + 1.4286(1 - 1.4286U) = 1,$$

$$2 - 4U + 1.4286 - 2.0408U = 1 \quad \Longrightarrow \quad 3.4286 - 6.0408U = 1,$$

$$U = \frac{2.4286}{6.0408} = 0.4020.$$

Back-substituting:

$$q_3 = \frac{1}{0.50}\left(1 - \frac{0.4020}{0.50}\right) = 2 \times 0.1960 = 0.3920,$$

$$q_4 = \frac{1}{0.70}\left(1 - \frac{0.4020}{0.70}\right) = 1.4286 \times 0.4257 = 0.6081.$$

The node investments are $x_3 = 1 - U/b_3 = 0.1960$ and $x_4 = 1 - U/b_4 = 0.4257$, with costs

$$\frac{x_3^2}{2} = 0.0192, \qquad \frac{x_4^2}{2} = 0.0906.$$

Edge-based equilibrium

In the edge-based model T_3 keeps $m_3 = 1$, but T_4 now has $m_4 = 2$. The first-target equation from Chapter 2, $q_i = m_i(b_i - U)/b_i^2$, applies to each:

$$q_3 = \frac{1 \cdot (0.50 - U)}{0.50^2}, \qquad q_4 = \frac{2 \cdot (0.70 - U)}{0.70^2}.$$

Imposing $q_3 + q_4 = 1$:

$$\frac{0.50 - U}{0.25} + \frac{2(0.70 - U)}{0.49} = 1.$$

Expanding,

$$4(0.50 - U) + 4.0816(0.70 - U) = 1,$$

$$2 - 4U + 2.8571 - 4.0816U = 1 \quad \Longrightarrow \quad 4.8571 - 8.0816U = 1,$$

$$U = \frac{3.8571}{8.0816} = 0.4773.$$

Back-substituting:

$$q_3 = \frac{0.50 - 0.4773}{0.25} = \frac{0.0227}{0.25} = 0.0909,$$

$$q_4 = \frac{2(0.70 - 0.4773)}{0.49} = \frac{2 \times 0.2227}{0.49} = 0.9091.$$

The per-edge investments are $x_{3,e} = 1 - U/b_3 = 1 - 0.9546 = 0.0455$ on T_3's single incoming edge, and $x_{4,e} = 1 - U/b_4 = 1 - 0.6818 = 0.3181$ on each of T_4's two incoming edges.

The defense costs are

$$\text{cost}(T_3) = \frac{0.0455^2}{2} = 0.0010, \qquad \text{cost}(T_4) = 2 \cdot \frac{0.3181^2}{2} = 0.1012.$$

The comparison

Moving from node to edge defense raises the attacker's payoff from $U = 0.4020$ to $U = 0.4773$, an increase of 18.7%. The driver is the multi-path factor $m_4 = 2$: T_4 must now defend two separate incoming edges. The attack probability on T_4 rises from 0.6081 to 0.9091, while T_3's investment collapses from 0.1960 to 0.0455 because, at the higher equilibrium value $U = 0.4773$, defending T_3 (value 0.50) is barely worthwhile.

A.3 Worked Calculation for the Pure-Strategy Example

Chapter 2 illustrates the pure Nash equilibrium proposition with the line ATK–T_1–T_2, values $b_1 = 0.10$ and $b_2 = 0.50$, and states two figures: the decentralized payoff $U = 0.25$ and the cooperative payoff $U \approx 0.196$. The decentralized figure follows in one line from the proposition; the cooperative figure requires the support iteration, and we give it in full here.

Decentralized: the pure equilibrium

The pure Nash equilibrium condition (i) is $b_2(1 - b_2) \geq b_1$. Substituting, $0.50 \times 0.50 = 0.25 \geq 0.10$, so the condition holds and the attacker attacks T_2 with probability one. The defender of T_2 faces a certain attack and invests $x_2 = b_2 = 0.50$ on its single incoming edge; T_1, never attacked, invests nothing. The attacker's payoff is

$$U = b_2(1 - x_2) = b_2(1 - b_2) = 0.50 \times 0.50 = 0.25.$$

Cooperative: the support iteration in full

Under cooperation a single coordinator chooses both edge investments to minimise total expected loss. The support is found by the same elimination procedure used for the other examples: start with all targets, solve, and drop any target whose attack probability comes out non-positive.

Round 1: candidate support $\{T_1, T_2\}$. With both targets attacked, T_1 sits at position 1 (a first target) and T_2 at position 2 (reached through T_1). The coordinator's first-order condition for the position-1 investment gives $x_1 = b_1 = 0.10$. The attacker's indifference between T_1 and T_2 requires the survival to T_2 to satisfy $b_2(1-x_1)(1-x_2) = b_1(1-x_1)$, that is $1 - x_2 = b_1/b_2 = 0.10/0.50 = 0.20$, so $x_2 = 0.80$. Solving the coordinator's system for the attack probabilities at these investments gives

$$q_1 = -0.778, \qquad q_2 = 8.889.$$

The attack probability q_1 is negative, which is infeasible: T_1 cannot be in the support. It is dropped.

Round 2: candidate support $\{T_2\}$. With T_1 removed, T_1 becomes a shield — the coordinator still defends position 1, because it lies on the route to T_2, but T_1 is no longer itself attacked. The first attacked position is now position 2.

The two edges in front of T_2 are not structurally symmetric, and their investments must be solved for separately. The entrance edge (ATK, T_1) has a single investor and contributes a survival factor $(1 - y_1)$; the interior edge (T_1, T_2) has two investors, each investing y_2, and contributes a survival factor $(1-y_2)^2$. Writing the coordinator's loss for this single-attacked-target case as

$$\mathcal{L} = b_2\,(1-y_1)(1-y_2)^2 \;+\; \frac{y_1^2}{2} \;+\; y_2^2,$$

where the cost term y_2^2 reflects the two investors on the interior edge, the two first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial y_1} = 0 \implies y_1 = b_2(1-y_2)^2,$$
$$\frac{\partial \mathcal{L}}{\partial y_2} = 0 \implies y_2 = b_2(1-y_1)(1-y_2).$$

This is a genuine two-equation system; it is not correct to assume the two investments are equal a priori. Solving it, the solution does turn out to satisfy $y_1 = y_2$ for these particular values, but that is an output of the system, not an assumption. Writing the common value y, both equations reduce to $y = 0.50\,(1-y)^2$, that is $y^2 - 4y + 1 = 0$, whose root in $(0, 1)$ is

$$y_1 = y_2 = 2 - \sqrt{3} \approx 0.2679.$$

The attacker's payoff is the survival through both edges times the value of T_2:

$$U = b_2\,(1-y_1)(1-y_2)^2 = 0.50 \times \left(\sqrt{3}-1\right)^3 \approx 0.196.$$

The attack probability is $q_2 = 1$, which is strictly positive, so the procedure terminates with support $\{T_2\}$.

Deviation check. It remains to confirm the attacker does not want to deviate to the shield T_1. Attacking T_1 would yield $(1-y_1)\,b_1 = (\sqrt{3}-1) \times 0.10 \approx 0.073$, which is below $U \approx 0.196$. The attacker has no profitable deviation, so the cooperative equilibrium is confirmed.

The comparison

The decentralized payoff is $U = 0.25$; the cooperative payoff is $U \approx 0.196$. The difference, about 0.054, is the amount by which coordination weakens the attacker on this network

— consistent with Bloch's general finding that the attacker is weakly worse off under centralized defense.

A.4 Cooperative Equilibrium on a Two-Branch Tree

The cooperative analysis in Chapter 2 was developed for line networks. The first network beyond a line on which the analysis still admits a clean treatment is a tree, in particular the simplest tree where two lines meet at the attacker. We give the worked calculation here, both to illustrate the decomposition that makes the case tractable and to identify precisely where the structure begins to break for more general networks.

Setup

Consider the network shown below: the attacker (node ATK) sits at the convergence point and two lines fan outward. The first branch has two targets A_1, A_2 with values $b_{A_1} = 0.20$ and $b_{A_2} = 0.50$. The second branch has two targets B_1, B_2 with values $b_{B_1} = 0.30$ and $b_{B_2} = 0.60$.

$$\begin{aligned} \text{ATK } –A_1–A_2 : &\quad \text{branch } A \\ \text{ATK } –B_1–B_2 : &\quad \text{branch } B \end{aligned}$$

An attack down branch A never crosses branch B and vice versa: the two branches share only the attacker node, which has no investment. This is the structural feature that makes the case tractable.

Identifying the support

The same elimination procedure used on a line applies independently to each branch. The two low-value targets A_1 and B_1 are dropped by the same logic that drops T_1 and T_2 in Example 2: their values are below the equilibrium U, so positive attack probabilities on them are infeasible. They become shields. The cooperative support is therefore $\{A_2, B_2\}$, with A_1 and B_1 defended but not attacked.

Solving the equilibrium: branch decomposition

With support $\{A_2, B_2\}$, there are four investments to determine: y_{e_A} on the entrance edge (ATK, A_1), y_{A_2} on the interior edge (A_1, A_2), y_{e_B} on the entrance edge (ATK, B_1), and y_{B_2} on the interior edge (B_1, B_2).

The key observation is that each branch's investments enter the loss function independently of the other branch. The two branches are coupled only through the attacker's common equilibrium payoff U: by Lemma 1, all attacked targets must yield the same value, so

$$b_{A_2}\,(1-y_{e_A})(1-y_{A_2})^2 \;=\; b_{B_2}\,(1-y_{e_B})(1-y_{B_2})^2 \;=\; U.$$

For any candidate value of U, each branch can be solved separately. The coordinator chooses the investments on that branch to minimise the branch's cost subject to the branch's indifference constraint. The remaining task is to identify the value of U for which the attack probabilities across both branches sum to one.

Branch A subproblem. For a given U, the coordinator minimises $y_{e_A}^2/2+y_{A_2}^2$ subject to $b_{A_2}(1-y_{e_A})(1-y_{A_2})^2=U$. The interior edge cost is $y_{A_2}^2$ rather than $y_{A_2}^2/2$ because two endpoints invest. The Lagrangian first-order conditions are

$$\begin{aligned} y_{e_A} &= \lambda_A\, b_{A_2}\,(1-y_{A_2})^2, \\ y_{A_2} &= \lambda_A\, b_{A_2}\,(1-y_{e_A})(1-y_{A_2}), \end{aligned}$$

where λ_A is the multiplier on the indifference constraint. Dividing the two equations yields $y_{e_A}(1-y_{e_A})=y_{A_2}(1-y_{A_2})$, whose solution in $(0,1)$ has $y_{e_A}=y_{A_2}$: the coordinator invests the same amount on the entrance edge and the interior shield edge of the branch. The same identity holds on branch B, giving $y_{e_B}=y_{B_2}$.

The coupling. The two branches share only the common U. The attack probability on each branch's attacked target is the corresponding Lagrange multiplier: $q_{A_2}=\lambda_A$ and $q_{B_2}=\lambda_B$. From the interior FOC on each branch,

$$q_{A_2}=\frac{y_{A_2}}{b_{A_2}\,(1-y_{e_A})(1-y_{A_2})}, \qquad q_{B_2}=\frac{y_{B_2}}{b_{B_2}\,(1-y_{e_B})(1-y_{B_2})}.$$

The equilibrium value of U is the one that makes $q_{A_2}+q_{B_2}=1$.

Numerical solution. Solving the constrained branch problems and adjusting U until the attack probabilities sum to one, the equilibrium is

$$U=0.2981,\quad y_{e_A}=y_{A_2}=0.1584,\quad y_{e_B}=y_{B_2}=0.2080,\quad q_{A_2}=0.4473,\quad q_{B_2}=0.5527.$$

The total cost is

$$C=\frac{y_{e_A}^2}{2}+y_{A_2}^2+\frac{y_{e_B}^2}{2}+y_{B_2}^2=0.1025.$$

Verifying the shield deviation conditions

The procedure terminates only if attacking either shield yields strictly less than U. The check passes on both branches: attacking A_1 would give $(1-y_{e_A})\,b_{A_1}=0.842\times 0.20\approx$

$0.168 < U = 0.298$, and attacking B_1 would give $(1-y_{e_B})\, b_{B_1} = 0.792 \times 0.30 \approx 0.238 < U$. The cooperative equilibrium is confirmed.

Why the analysis generalises to any two-branch tree, and where it stops

The decomposition above works because the two branches are structurally independent: any investment on branch A affects only attacks that travel down branch A, and likewise for branch B. The branches couple only through the attacker's equilibrium payoff U. The same logic extends to two-branch trees of any branch length: each branch's investments are determined by its deepest attacked target, the common U, and the branch's own indifference constraint, while U is pinned down by the cross-branch requirement $\sum q = 1$.

The decomposition stops working as soon as a single target is reached through multiple paths in the network interior. On the general network of Example 2, target T_4 is reached through both T_1 and T_2, and the entrance edge (ATK, T_1) carries attacks heading to both T_3 and T_4. The coordinator's first-order condition on $y_{(\text{ATK},T_1)}$ then couples this single investment to two different downstream targets simultaneously — one term for the T_3 path, one for the T_4 path. The FOCs no longer separate into independent branch subproblems, and a clean analytical characterisation analogous to the one above is no longer available. The full system of five investments, three indifference constraints, and three path-level attack probabilities can still be set up and solved numerically, but the structural simplicity that makes the line and the two-branch tree tractable is lost.

A.5 Worked Calculation for the Comparative Experiment of Chapter 4

Chapter 4 reports an experiment in which a new connection is added to the general network of Example 2: an edge is built between T_3 and T_4. We give the calculation here.

Effect of the new edge

Before the new edge, the edge-based equilibrium of Example 2 has support $\{T_3, T_4\}$ with $m_4 = 2$ and $U = 0.4773$. Adding the edge (T_3, T_4) creates a new route to T_4: the attacker can travel ATK $\to T_1 \to T_3 \to T_4$. In the modified equilibrium T_3 turns out to drop from the support, so the nodes T_1 and T_3 both invest zero, and the path ATK $\to T_1 \to T_3 \to T_4$ becomes a third free-highway edge into T_4. The multi-path factor of T_4 rises from $m_4 = 2$ to $m_4 = 3$.

Re-solving the equilibrium

With T_3 out of the support, T_4 is the only attacked target, and the support is $\Delta = \{T_4\}$ with $m_4 = 3$. The single equilibrium condition is $q_4 = 1$, with

$$q_4 = \frac{m_4(b_4 - U)}{b_4^2} = \frac{3(0.70 - U)}{0.49} = 1.$$

Solving,

$$3(0.70 - U) = 0.49 \implies 0.70 - U = 0.1633 \implies U = 0.5367.$$

The per-edge investment on each of T_4's three incoming edges is

$$x_{4,e} = 1 - \frac{U}{b_4} = 1 - \frac{0.5367}{0.70} = 1 - 0.7667 = 0.2333.$$

Interpretation

Adding the edge raises the attacker's payoff from $U = 0.4773$ to $U = 0.5367$, an increase of about 12.4%. Target T_3, which had a small positive attack probability of 0.0909 before the modification, drops out of the support entirely: all attack probability now concentrates on T_4. This is the logical chain of Chapter 2 in action — an additional free-highway edge raises m_4, which raises U, which pushes the lower-value target T_3 off the margin. It also confirms that the edge-based model exhibits no Braess paradox: adding a connection strictly benefits the attacker.

A.6 Algebra for the Cost-Savings Inequality

The proof of the cost-savings result in Chapter 2 reduces the comparison, at each interior position of a line, to showing that

$$\frac{(1-u)^2}{2} - (1-\sqrt{u})^2 > 0 \qquad \text{for all } u \in (0,1),$$

where $u = 1 - y$ and y is the node-based investment at that position. We verify the inequality here.

Define

$$g(u) = \frac{(1-u)^2}{2} - (1-\sqrt{u})^2.$$

Expanding both terms:

$$\frac{(1-u)^2}{2} = \frac{1-2u+u^2}{2} = \frac{1}{2} - u + \frac{u^2}{2},$$

$$(1-\sqrt{u})^2 = 1 - 2\sqrt{u} + u.$$

Subtracting,

$$g(u) = \left(\frac{1}{2} - u + \frac{u^2}{2}\right) - \big(1 - 2\sqrt{u} + u\big) = -\frac{1}{2} - 2u + \frac{u^2}{2} + 2\sqrt{u}.$$

Substitute $u = t^2$ with $t = \sqrt{u} \in (0,1)$:

$$g = -\frac{1}{2} - 2t^2 + \frac{t^4}{2} + 2t = \frac{t^4 - 4t^2 + 4t - 1}{2}.$$

The quartic in the numerator, $h(t) = t^4 - 4t^2 + 4t - 1$, has $h(1) = 1 - 4 + 4 - 1 = 0$, so $(t-1)$ is a factor. Dividing,

$$h(t) = (t-1)(t^3 + t^2 - 3t + 1).$$

The cubic factor has $t = 1$ as a root as well: $1 + 1 - 3 + 1 = 0$. Dividing again,

$$t^3 + t^2 - 3t + 1 = (t-1)(t^2 + 2t - 1).$$

Therefore

$$h(t) = (t-1)^2(t^2 + 2t - 1).$$

For $t \in (0,1)$, the factor $(t-1)^2$ is strictly positive. The remaining factor $t^2 + 2t - 1$ is increasing on $(0,1)$ and has its single positive root at $t = \sqrt{2} - 1 \approx 0.414$. So $h(t) < 0$ for $t < \sqrt{2} - 1$ and $h(t) > 0$ for $t > \sqrt{2} - 1$.

This means $g(u)$ changes sign once on $(0,1)$, at $u = (\sqrt{2}-1)^2 = 3 - 2\sqrt{2} \approx 0.172$. The cost-savings inequality $g(u) > 0$ therefore holds for $u > 3 - 2\sqrt{2}$, that is, for node-based investment $y = 1 - u$ below $1 - (3 - 2\sqrt{2}) = 2\sqrt{2} - 2 \approx 0.828$.

In every equilibrium that arises in the model, the node-based investment at an interior position is well below this threshold — equilibrium investments are bounded by the target values, which lie in $(0,1)$, and interior cooperative investments in the worked examples are an order of magnitude smaller than 0.828. Within the relevant range the inequality holds strictly, and the cost saving is therefore strictly positive at every interior position, as claimed in Chapter 2. At the numerical values of Example 1, where $y = 0.2441$ so $u = 0.7559$, we have $g(u) > 0$ directly: the node-based per-position cost 0.0298 exceeds the edge-based per-position cost 0.0171.

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